FORCASTING THE AUSTRALIAN YIELD CURVE

Group M

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1.0 Abstract

Our paper develops a macro-finance framework to forecast Australia's AGB yield curve under structural shocks. We estimate a dynamic Nelson–Siegel model in state-space form using the Kalman filter, with monthly RBA zero coupon yields anchoring the measurement equation (section 5). The extracted level, slope and curvature factors are then linked to key macroeconomic and financial variables in a VECM(1) system (section 6). This framework then allows us to conduct scenario analysis on three stylised shocks: a rise in debt-to-GDP, a sharp fall in oil prices and a downward productivity revision (section 3). Each provides distinctive yield-curve responses, consistent with theory: bear steepening under debt issuance, front-end compression under disinflation and moderate flattening from productivity decline (section 7). Results offer policy relevance for debt issuance timing and monetary policy communication (section 8). Our findings suggest that macro-fiscal shocks propagate differentially along the term structure, underscoring the need for scenario-based tools in yield forecasting and public-finance planning for the AOFM and RBA.

2.0 Introduction

Forecasting the term structure of interest rates is critical for economic policy, financial markets, institutional risk management, and everyday Australians. As a compressed reflection of future interest rate expectations, inflation dynamics, and broader macroeconomic conditions, the yield curve is not merely a passive indicator; it is an essential input into decisions made by central banks, governments, credit agencies, financial institutions, and households.

At the macro level, the shape of the yield curve provides insights into the expected trajectory of the economy. For example, an inverted yield curve (where short-term rates exceed long-term ones) has historically preceded economic downturns, making it a useful recession predictor (Estrella & Mishkin, 1996). Central banks, such as the Reserve Bank of Australia (RBA), closely monitor the curve to assess whether markets expect rate hikes or cuts, which helps in calibrating monetary policy settings (Alim & Connolly, 2018). In Australia's context, changes to the slope and level of the curve, such as bear steepening due to increased long-term debt issuance or bull flattening from disinflationary oil shocks, can materially affect the RBA's communication strategy and policy stance (RBA, 2024).

Yield curve forecasts guide financial market participants in pricing government bonds, credit instruments, and derivatives. Fixed-income portfolio managers use expected curve shifts to structure carry-trade strategies, duration positioning, and hedge macro risk. For banks and insurers, the curve determines lending spreads, asset-liability duration matching, and valuation of long-dated liabilities. Importantly, Australian government bonds are used as benchmarks to price corporate debt, infrastructure projects, and public-private financing. An upward shift in long-term yields due to rising debt or productivity constraints, as projected under current PEFO assumptions (Australian Treasury, 2025), would directly increase these financing costs, influencing capital allocation decisions across the economy.

For sovereign risk analysts and credit rating agencies like Moody's, the yield curve offers real-time feedback on a country's fiscal sustainability and market credibility. A steepening yield curve amid stable short rates may indicate rising term premia or fiscal management concerns, particularly if the curve shift is driven by increased long-dated government bond supply (AOFM, 2025). Forecasting how the curve responds to debt issuance helps governments plan more efficient funding strategies and mitigate rollover or interest-rate risks. As Moody's noted in its most recent outlook, structural productivity constraints and expanding debt levels could influence the cost of sovereign borrowing over the long term (Moody's Investors Service, 2025).

Beyond institutional implications, yield curve forecasts are directly relevant to Australian households and businesses. For example, fixed mortgage rates are often priced off medium- to long-term bond yields. A steepening yield curve, such as that driven by the AOFM's shift to long-dated issuance (Australian Treasury, 2025), increases these borrowing costs even if the RBA leaves the cash rate unchanged. Similarly, businesses issuing corporate debt or investing in capital projects rely on long-term interest rate expectations when evaluating investment viability. Therefore, understanding and forecasting the curve provides foresight into future credit conditions, housing affordability, and cost-of-capital dynamics, allowing for more informed decision-making at all levels of the economy.

3.0 Background

We forecast the impact of several plausible scenarios on the level, slope, and curvature of the Australian yield curve in one year. Our model assumes the three key shocks through a meta-analysis (research) approach: (i) a sharp decline in oil prices; (ii) a rise in the debt to GDP ratio; (iii) a downgrade to long-run productivity estimates. Grounded in current economic concerns and projections, these scenarios have clear implications for households and investors, via their effects on inflation, interest rates and bond yields, and for credit rating agencies through the outlook for Australia's fiscal health and growth potential.

Oil prices

Global oil production in 2026 is forecast to outpace demand growth, pushing prices down to USD $50pb, consistent with Goldman Sachs, JP Morgan, and EIA projections *(figure 1)* (Slav, I. 2025; J.P. Morgan Global Research. 2025; EIA. 2025). While demand is expected to rise by 1.4m bpd from upward revisions to economic growth forecasts to 3.1%, this is outpaced by OPEC's planned 2.5m bpd production increase, creating a surplus of roughly 3m bpd (Reuters 2025). This ramp-up is primarily driven by Saudi Arabia's response to "cheating" member states – notably Kazakhstan, which overshot production beyond OPEC+ quota to increase market share. Additionally, Trump is expected to continue pressuring OPEC to deliver low prices to suppress Russian export revenues in hopes of neutralising its threat to Ukraine, and curbing tariff-related inflation as reinstated by JP Morgan's Kaneva: *"We don't anticipate the administration intervening to stabilize prices unless crude falls below $50"*. However, even if geopolitical tensions ease and a ceasefire prevails between Russia and Ukraine, the removal of sanctions on Russian oil would boost global supply and unwind the risk premium in crude, making this scenario more palpable.

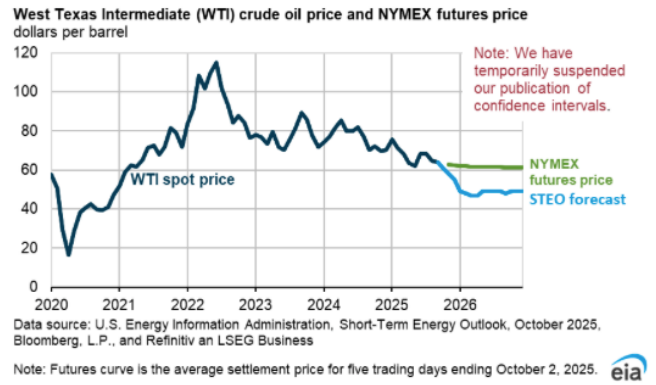


Figure 1

Debt to GDP

Treasury's Pre-election Economic and Fiscal Outlook projects gross debt to GDP to rise from 35.5% ($1,022bn) in 2025, to 36.5% ($1,092b) over 2026 as additional borrowings are required to neutralise an underlying cash deficit of $35.4b (arising from govt payments of $801.3b exceeding receipts of $766.0b) *(figure 2)*. Key drivers include cost-of-living relief measures in Medicare, education, and childcare, alongside a $22.7 billion stimulus over the decade for the *Future Made in Australia* agenda. On the revenue side, personal income tax cuts commencing 1 July 2026 lower the 16% marginal rate on incomes between $18,201–$45,000 to 15%, reducing receipts by $17.1 billion over five years. To finance this shortfall, AOFM skews bond issuances to longer term securities *(figure 3),* thereby extending payment horizons and mitigating refinancing risk. Correspondingly, these securities incorporate a term premium to compensate investors for macroeconomic uncertainty, elevating the long end of the yield curve

to create a bear steepening trend. Notably, PEFO gross debt projections for 2026 have remained unchanged year-on-year, implying the rise in debt stems from structural deficits rather than new policy shocks. Its persistence makes the scenario highly probable and suitable for analysis.

Figure 3

Table 3: Australian Government general government sector budget aggregates

	Estimates						Projections
	2024-25	2025-26	2026-27	2027-28	2028-29	Total	2035-36
	$b	$b	$b	$b	$b	$b	% of GDP
Underlying cash balance	**-27.9**	**-42.2**	**-35.4**	**-37.1**	**-37.0**	**-179.5**	
Per cent of GDP	-1.0	-1.5	-1.2	-1.2	-1.1		0.0
Receipts	**703.9**	**735.4**	**766.0**	**797.4**	**840.8**	**3,843.5**	
Per cent of GDP	25.3	25.5	25.6	25.3	25.3		26.8
Tax receipts	**645.2**	**676.1**	**707.6**	**735.9**	**778.3**	**3,543.1**	
Per cent of GDP	23.1	23.5	23.6	23.4	23.4		24.9
Non-tax receipts	**58.7**	**59.3**	**58.4**	**61.4**	**62.6**	**300.4**	
Per cent of GDP	2.1	2.1	2.0	2.0	1.9		1.9
Payments(a)	**731.8**	**777.5**	**801.3**	**834.5**	**877.9**	**4,023.0**	
Per cent of GDP	26.3	27.0	26.8	26.5	26.4		26.8
Gross debt(b)	**940.0**	**1,022.0**	**1,092.0**	**1,161.0**	**1,223.0**		
Per cent of GDP	33.7	35.5	36.5	36.9	36.8		31.9
Net debt(c)	**556.3**	**620.7**	**676.3**	**713.9**	**768.2**		
Per cent of GDP	20.0	21.6	22.6	22.7	23.1		20.2
Net interest payments(d)	**14.9**	**18.5**	**21.2**	**26.9**	**28.1**	**109.6**	
Per cent of GDP	0.5	0.6	0.7	0.9	0.8		1.1

a) Equivalent to cash payments for operating activities, purchases of non-financial assets and principal payments of lease liabilities.

b) Gross debt measures the face value of Australian Government Securities (AGS) on issue and is presented as at the end of the financial year. The change in the underlying cash balance since the 2025–26 Budget has not materially changed the Government's financing requirement. Gross debt estimates therefore remain unchanged since the 2025–26 Budget.

c) Net debt is the sum of interest-bearing liabilities (which includes AGS on issue measured at market value) less the sum of selected financial assets (cash and deposits, advances paid and investments, loans and placements) and is presented as at the end of the financial year.

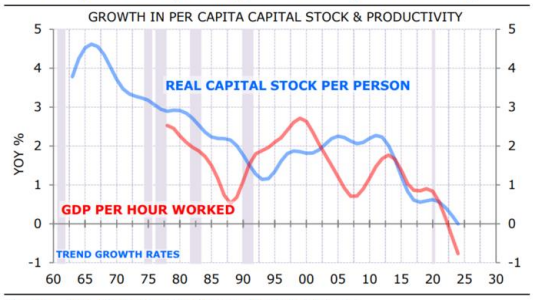
d) Net interest payments are equal to the difference between interest payments and interest receipts.

Figure 2

Productivity growth

The RBA's August 'Statement of Monetary Policy' revealed that persistently weaker growth in capital deepening and multifactor productivity led to a downward revision in its long run productivity growth forecast from 1% to 0.7%. Morgan Stanley's Gerard Minack predicates slower growth in capital deepening on rising population growth, with firms lacking sufficient resources to maintain strong investment per worker (*figure 4*). Moreover, the Productivity Commission finds that merely 2% of Australian firms innovate in ways that are 'new to the world' - with 98% relying on the adoption of dated technologies which are cumbersome and difficult to integrate with existing factors of production - imposing a structural barrier to productivity growth. Lower productivity constrains potential output, causing demand to exceed supply, bidding up prices and driving long run inflationary pressures. Correspondingly, the RBA will likely increase longer run neutral interest rates to curtail excess demand, driving long run yields higher to create a bear steepening scenario.

Less capital deepening = slower productivity



Source: ABS, Melbourne Institute; Minack Advisors

Figure 4

4.0 Literature Review

The dynamics of interest rates lie at the heart of macroeconomic stability, financial market functioning, and policy transmission. Understanding the term structure of interest rates, the relationship between yields and maturities, has long been a priority for economists and policymakers, as it captures how markets price risk, growth expectations and inflation over time. Despite decades of research, the challenge persists: how can we model the evolution of the yield curve in a way that captures both its empirical regularities and its deep structural drivers in the macroeconomy? Recent periods of heightened uncertainty, marked by unconventional monetary policy, global financial shocks, and shifting inflation expectations, have reignited interest in this question.

Existing models often fall into two extremes: those that treat yield curves as purely statistical constructs and those that impose rigid theoretical constraints that limit empirical flexibility. This leaves a gap for approaches that combine the dynamic adaptability of data-driven models with the interpretability of macroeconomic structure.

Our study aims to address this gap by exploring the dynamic interactions between the yield curve and key macroeconomic variables through a Vector Error Correction Model (VECM) and Structural Vector Autoregression (SVAR) framework. By embedding the Nelson–Siegel factor structure into a macro econometric system, we investigate how long-term equilibrium relationships and short-run adjustments shape interest rate dynamics. This builds upon three pivotal strands of literature that inform our methodological and conceptual approach: the dynamic yield curve framework developed by Diebold and Rudebusch (2013), the integration of macroeconomic fundamentals by Diebold, Rudebusch and Aruoba (2006), and the structural identification principles introduced by Blanchard and Quah (1989).

Diebold and Rudebusch's *Yield Curve Modeling and Forecasting: The Dynamic Nelson–Siegel Approach* (2013) represents a cornerstone in modern yield curve analysis. Their work introduced the Dynamic Nelson–Siegel (DNS) model, which captures the shape of the yield curve through three latent factors, level, slope, and curvature, each corresponding to distinct maturity movements. This simple factorisation offers a powerful empirical representation of yield dynamics while maintaining interpretability in economic terms: the level relates to long-term expectations, the slope reflects monetary policy and short-term rates, and curvature captures medium-term fluctuations The significance of the DNS model lies in its dynamic and state-space representation, allowing for the application of Kalman filtering and forecasting techniques. This framework advances beyond static term structure models by recognising the persistence and co-movement of yield factors over time. Importantly, Diebold and Rudebusch demonstrate that despite its simplicity, the model rivals more complex no-arbitrage specifications in forecasting performance, particularly for medium horizons.

For our study, their contribution serves as the methodological foundation. The Nelson–Siegel decomposition enables us to distil the information embedded in the yield curve into a tractable set of dynamic variables. These factors then become the basis for our VECM analysis, where we examine how they co-evolve with macroeconomic indicators. Furthermore, the discussions in Diebold and Rudebusch (2013) on the balance between empirical flexibility and theoretical constraints inform our decision not to impose full arbitrage-free restrictions, prioritising empirical adaptability and the integration of broader economic variables.

While the DNS framework elegantly captures yield dynamics, it does not inherently explain their macroeconomic underpinnings. Diebold, Rudebusch and Aruoba's (2006) paper, *The Macroeconomy and the Yield Curve: A Dynamic Latent Factor Approach*, fills this gap by linking the yield curve directly to macroeconomic variables. Their dynamic factor model jointly estimates latent yield factors and observable macro variables such as output, inflation, and monetary policy rates. This allows them to capture bi-directional interactions: macro conditions influence yield movements, and yield curves, in turn, signal future economic activity. Their key finding, that including macroeconomic information improves yield forecasting, redefined how researchers approach term structure modelling. It

demonstrated that the yield curve is not merely a reflection of market expectations but also a channel through which monetary and real economy signals propagate. Moreover, their model revealed that macro shocks often manifest through predictable movements in the yield factors: for instance, contractionary monetary policy typically flattens the slope and raises the level of the yield curve.

In our work, this insight provides the conceptual bridge between financial and macroeconomic dynamics. We extend the spirit of their model using a VECM framework, which captures both long-run cointegrating relationships and short-run dynamics between the Nelson–Siegel factors and macro variables such as inflation, GDP growth and unemployment. By embedding these interactions within a cointegration structure, we can assess whether yield factors share stable equilibrium relationships with the broader economy, a question central to understanding monetary transmission and expectations formation.

While Diebold and Rudebusch's frameworks focus on capturing dynamic co-movements, they are largely reduced form in nature. To understand causal mechanisms, how specific shocks propagate through the system, structural identification is required. Here, Blanchard and Quah's (1989) seminal paper, *The Dynamic Effects of Aggregate Demand and Supply Disturbances*, provides the foundation. They introduced a structural VAR approach that distinguishes between demand and supply shocks based on long-run restrictions: demand shocks have no permanent effect on output, whereas supply shocks do. This identification strategy revolutionised empirical macroeconomics, offering a transparent way to decompose observed fluctuations into meaningful structural components.

In our context, it informs the design of our SVAR extension, which allows us to trace how shocks, such as monetary tightening or inflation surprises, affect the yield curve's shape and macro variables over time. By combining the Blanchard–Quah methodology with the yield curve–macro framework, we move beyond correlation to explore the causal transmission channels embedded in the term structure.

Taken together, these three works delineate the intellectual trajectory that our study builds upon. Diebold and Rudebusch (2013) establish the empirical structure of the yield curve; Diebold, Rudebusch and Aruoba (2006) integrate this with macroeconomic fundamentals; and Blanchard and Quah (1989) provide the identification strategy for uncovering causal dynamics. Each fills a distinct layer, measurement, interaction, and interpretation, that collectively informs our modelling approach.

Our contribution lies in synthesising these frameworks within a modern econometric setting. By employing a VECM structure with structural restrictions, we capture both equilibrium relationships and dynamic adjustments between yield and macro factors. This approach extends prior work in three ways:

1. It adapts the Nelson–Siegel structure to a system where yield and macro variables are jointly endogenous.

2. It applies cointegration and error-correction principles to model long-run dependencies that earlier studies largely assumed rather than tested.

3. It incorporates structural identification to distinguish between shocks of different economic origins, providing richer policy insights.

In essence, while prior research has mapped the contours of the yield curve and its connection to the macroeconomy, the integration of these strands into a unified VECM–SVAR framework remains limited, particularly in the context of emerging economic conditions and evolving monetary policy regimes. By combining the dynamic, empirical strengths of the DNS model with the structural interpretability of SVAR, our study aims to provide a more holistic understanding of how macro-financial forces shape interest rate dynamics over time.

5.0 Theoretical Model

5.1 Nelson-Siegel equation for fitting the curve yield

The yield curve is characterised by three principal components: Level (Lt), Slope (St), and Curvature (Ct), extracted via the Nelson-Siegel (1987) parametric form, which imposes a functional structure to directly estimate these LSC factors:

$$y_t(\tau) = L_t + S_t F_1(\tau) + C_t F_2(\tau) + \varepsilon_t, \text{ or as:}$$

$$y_t(\tau) = \beta_0 + \beta_1\left(\frac{1-e^{-\lambda_t \times \tau}}{\lambda_t \times \tau}\right) + \beta_2\left(\frac{1-e^{-\lambda_t \times \tau}}{\lambda_t \times \tau} - e^{-\lambda_t \times \tau}\right) + \varepsilon_t$$

$$y_t(\tau) = \text{zero coupon yield at time } t \text{ for maturity } \tau$$

$$\tau = [0year,\ 0.25\ year\ldots, 10year]$$

$$\beta_0 = L_t = \text{level factor}$$

$$\beta_1 = S_t = \text{slope factor}$$

$$\beta_2 = C_t = \text{curvature factor}$$

$$\lambda_t = \text{decay parameter}$$

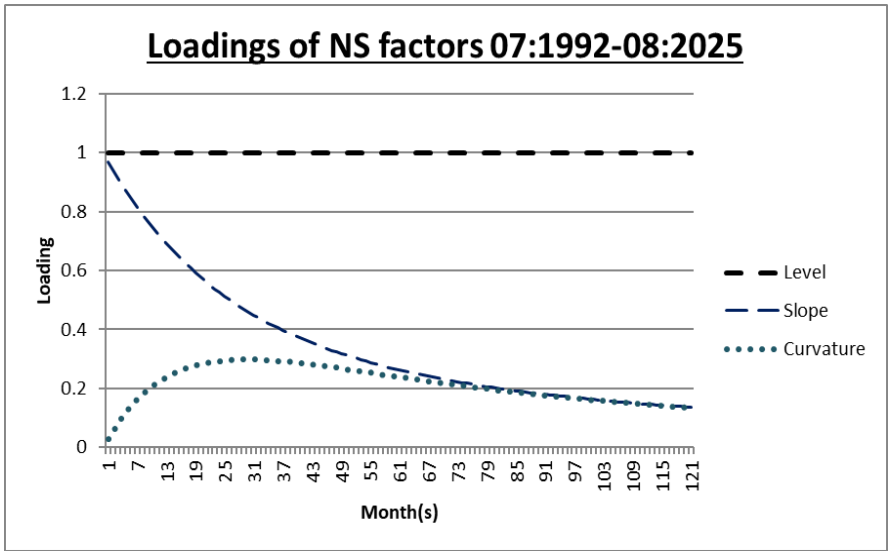
$$\varepsilon_t = \text{error term to improve empirical fit}$$

Following Nymand-Andersen (2018), we use a panel of zero-coupon (spot) yields across maturities, where, to avoid long estimation lead times, we utilise existing data from the RBA Data Hub (n.d.) at a monthly frequency (read τ as months). The measurement equation then treats the zero-yield vector each date as a noisy linear combination of Level, Slope, and Curvature. This choice does not hinder comparisons with coupon bonds, as coupon-bond yields (par/YTM) are nonlinear transforms of the same discount factors and need not equal spot rates, with prices invariant once cash flows are discounted using the zero curve (Bank for International Settlements, 2005).

5.2 Interpretation of factors

We avoid static PCA because it is a statistical atheoretical rotation, with factors orthogonal by construction and sample-dependent (sign/scale indeterminacy, loading drift). By contrast, a state-space Nelson-Siegel model delivers persistent, interpretable Level, Slope, Curvature factors with an explicit (e.g., AR (1)) transition equation, where a non-diagonal state-shock covariance Q permits contemporaneous co-movement among factor innovations, with Kalman filtering naturally handling measurement error (Durbin & Koopman, 2012).

The level factor shifts the whole curve up or down by roughly the same amount and typically explains the majority share of variation (around 80%). Its loading is constant across maturities, so every yield moves together when level moves. The slope factor tilts the curve; short rates react more than long rates and commonly accounts for about 15% of variation. Its loading starts near one at very short maturities and decays toward zero as maturity lengthens. The curvature factor shapes the middle of the curve (a hump), with limited movement at the very short and very long ends and tends to explain the residual share (about 5%) (Litterman and Scheinkman, 1991). In this framework, policy moves are transmitted mainly at the front end, while the long end reflects expectations and term-premia, relationships the state-space model can represent directly through its transition and measurement equations, which PCA cannot.



5.3 State space representation of the factor model

Because the Diebold-Li model has parametric form characteristics that are impossible to specify explicitly, this specification (5.1) creates a state-space object implicitly. For estimation purposes, we assume that the factors follow a vector autoregressive process of order 1, where any AR/MA process may be written in the state space form (Hamilton, 1994). Each factor in the Diebold-Li model includes a nonzero offset (mean), which represents a regression component, where equations in Eviews follow:

Measurement equation

$y_t(\tau) = Z(\lambda)s_t + \varepsilon_t$, *where* $\varepsilon_t \sim (0, H_t)$. Refer also to section 5.1.

$$\begin{bmatrix} y_{1,t} \\ y_{N,t} \end{bmatrix} = \begin{bmatrix} 1 & \frac{1-e^{-\lambda\tau}}{\tau_1/\lambda} & \frac{1-e^{-\lambda\tau}}{\tau_1/\lambda} - e^{-\lambda\tau} \\ 1 & \frac{1-e^{-\lambda\tau}}{\tau_N/\lambda} & \frac{1-e^{-\lambda\tau}}{\tau_N/\lambda} - e^{-\lambda\tau} \end{bmatrix} \begin{bmatrix} L_t \\ S_t \\ C_t \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,t} \\ \varepsilon_{N,t} \end{bmatrix}$$

Transition equation

$s_{t+1} = Ts_t + \eta_t$, *where* $\eta_t \sim (0, Q)$

Which reduces to:

$s_{t+1} - \mu = T(s_t - \mu) + \eta_t$

$$\begin{bmatrix} L_{t+1} - \mu_L \\ S_{t+1} - \mu_S \\ C_{t+1} - \mu_C \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} L_{t-1} - \mu_L \\ S_{t-1} - \mu_S \\ C_{t-1} - \mu_C \end{bmatrix} + \begin{bmatrix} \eta_t(L) \\ \eta_t(S) \\ \eta_t(C) \end{bmatrix}$$

Here, measurement errors capture yield movements not explained by the factors, where we assume these are orthogonal to state shocks and uncorrelated across maturities (diagonal H). By contrast, we allow factor innovations to co-move via a non-diagonal Q.

$$\begin{bmatrix} \eta_t \\ \varepsilon_t \end{bmatrix} \sim N\left[\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} Q & 0 \\ 0 & H \end{bmatrix}\right]$$

Estimating all unknown measurement factors under MLE, we further allow lambda to be freely estimated across each observation using the Kalman filter (see appendix 11.1.2), which sets the maturity-decay profile of the factor loadings (a slower decay with smaller lambda). However, in sections 5-6, we follow Diebold and Li (2006), fixing lambda to place the curvature "hump" at around 30 months in order stabilise identification and keep results comparable across studies for current period forecasting specifications.

For a breakdown of the code in which the state-space framework is defined in Eviews, Eviews product guides may be consulted, as well as this example in (Matlab) which we extend with our 3 policy salient variables to compress coefficients on the yield factors (see next section, but note also that this example treats lambda as a time invariant parameter). Yield factor output values (L,S,C,λ) saved in xlsx. format can also be provided upon request.

5.4 Estimation output

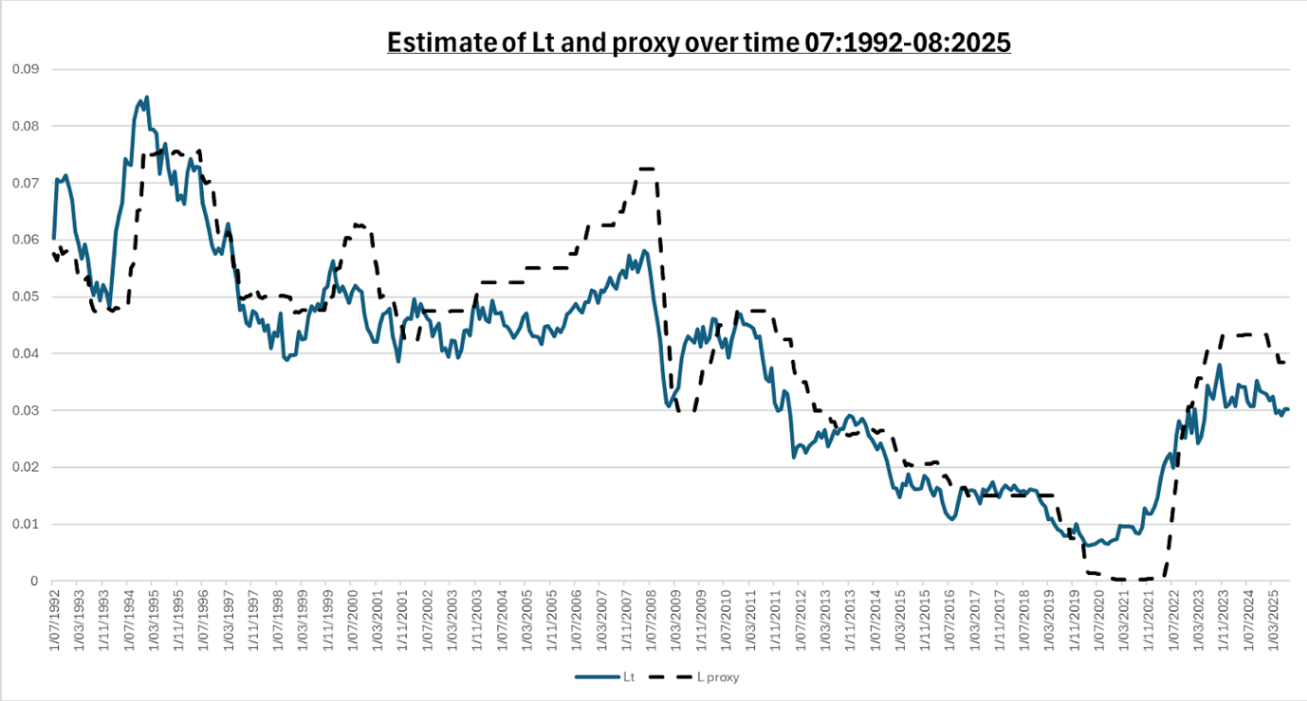
Following Nymand-Andersen (2018), as the yields-only model fits the yield curve data well, but the yields-macro model performs better, to study macro-term-structure links, we augment the system with a small set of policy-salient observed variables, including; the target cash rate (CSHR), inflation (proxy by trimmed mean, TMINF) and industry capacity utilisation (UTI) as a real-activity proxy. As the variables are perfectly observed, we ensure that their estimated measurement errors are close to zero (see appendix 11.1.1). The re-estimated model lets us quantify how much of the variation in the level, slope and curvature is attributed to policy, prices and activity.

Transition Matrix

$$\begin{bmatrix} L_{t+1} - \mu_L \\ S_{t+1} - \mu_S \\ C_{t+1} - \mu_C \\ UTI_{t+1} - \mu_{UTI} \\ CSHR_{t+1} - \mu_{CSHR} \\ TMINF_{t+1} - \mu_{TMINF} \end{bmatrix} = \begin{bmatrix} 0.9037 & 0.0975 & 0.0566 & -0.05517 & -0.1138 & 0.04956 \\ 0.2358 & 0.9226 & 0.0536 & 0.08608 & -0.0301 & -0.0789 \\ 0.0676 & -0.1781 & 0.8335 & 0.0682 & 0.3953 & -0.3370 \\ 0.8053 & 0.5805 & 0.0279 & 0.8417 & -1.037 & 0.1406 \\ 0.3126 & 0.2593 & 0.0078 & -0.0466 & 0.8889 & 0.0913 \\ 0.0925 & 0.1503 & 0.0531 & -0.0106 & -0.1938 & 1.0419 \end{bmatrix}$$

The yield factors are persistent (high AR effects), especially L (0.90) and S (0.92), with C driven mostly by its own lag (0.83). S responds to L (0.24), while L is only weakly affected by S and C. Here it is worth noting that mechanically, adding precise macro equations and the policy salient variables increases the information matrix and reduces the posterior variance of the level, slope and curvature, limiting the likelihood of aliasing biases. Accordingly, we exclude these macro series from the VECM forecasting system, which is reserved for modelling slow-moving, cointegrating macro fundamentals with interpretable equilibrium links to the yield curve. This two-stage separation, state-space for efficient factor extraction/decomposition, VECM for long-run forecast dynamics, is expected to provide robust estimation results without distorting the structural propagation of shocks used in our scenario analysis.

<u>Estimated factors versus their empirical proxies</u>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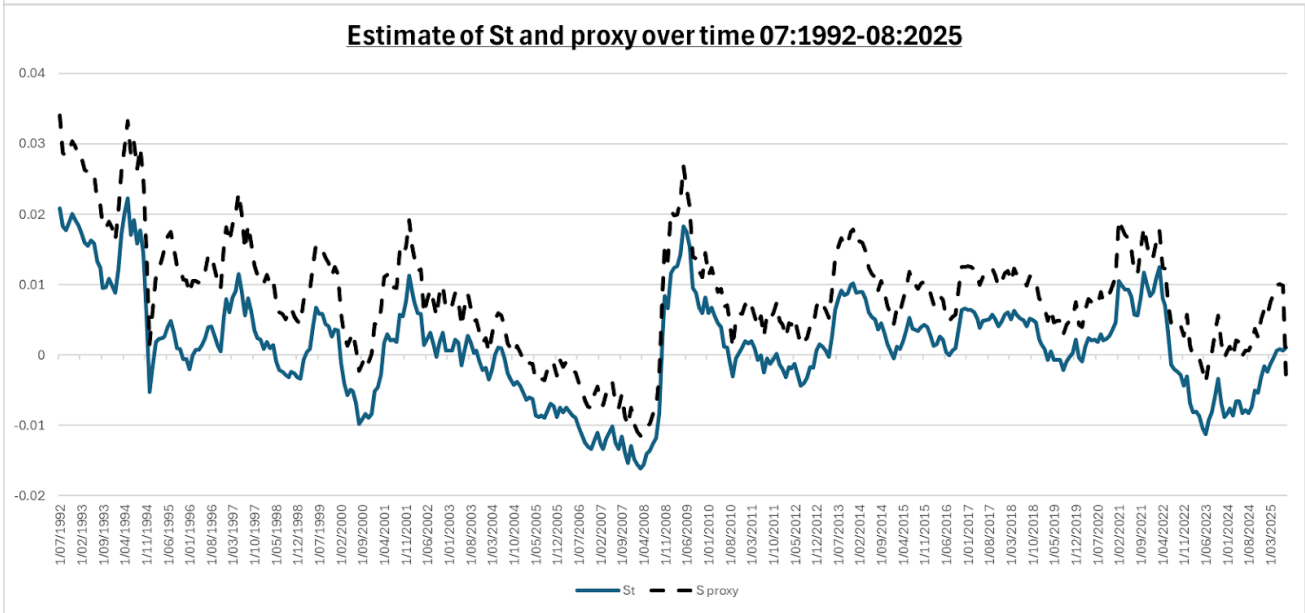
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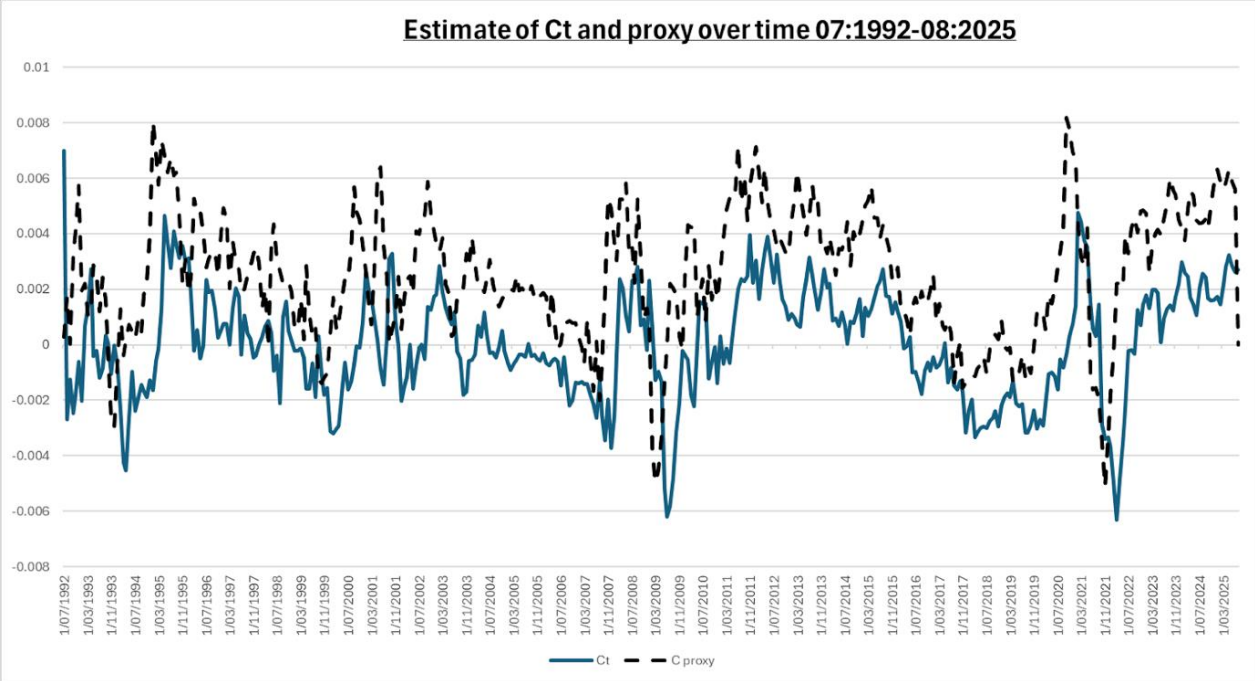


5.5 Variable selection theory

We retain monthly frequency in variables as it preserves signalling, mitigates over-parameterisation risk, and avoids temporal-aggregation bias. First, with k variables and p lags, VECM parameters grow roughly with *k^2p,* where moving to quarterly would slash the sample size by ~⅔, forcing very low p and inflating small-sample bias in Johansen tests and the α, Γ estimates. Monthly specification thus provides three times the observations, so we can keep a parsimonious lag length while maintaining degrees of freedom, reducing (not increasing) overfitting risk, given 12 system variables are retained (section 5.6).

OILP and yields also move at high frequency, so aggregating to quarters may dilute near-term IRFs. For DGDP (Δ(AGB stock/GDP)), the GDP denominator is sticky intra-quarter (1-3 months), but the AGB stock is a floating, market-linked series that exhibits meaningful MoM variation from issuance, buybacks, and maturities. In this way, a monthly frequency keeps that supply signal, which is precisely what links debt dynamics to the yield-curve level in our cointegrating space. Finally, issuance calendars and market pricing for AGBs are operationally monthly, so aligning to this data frequency is expected to improve timing consistency between shocks and responses.

Macro block |

The macroeconomic variables, PROD (industrial productivity), UNE (unemployment rate), INFGAP (inflation gap from expected), DGDP (Government debt/GDP ratio), and OILP (crude oil prices), primarily anchor the level of the yield curve. These variables are selected based on their long-run equilibrium relationships with interest rates, as output and GDP capture cyclical demand-side pressures, unemployment and the inflation gap reflect the degree of slack in the economy and central bank response, and oil prices act as a proxy for inflationary and trade pressures through external income shocks in the AS-AD framework. To maintain parsimony while retaining open-economy realism, we incorporate exchange rate dynamics via only the AUD/USD rate, based on sizeable trade volumes (AOFM, 2025). This choice is grounded in covered and uncovered interest rate parity (CIP and UIP) theory, which links yield differentials between Australia and the US to expected currency movements. Specifically, under UIP, the forward-looking nature of the AUD/USD spot rate embeds market expectations about relative monetary policy stances and macroeconomic outlooks, which may proxy for international capital flows and return differentials.

Financial block |

To enhance the model's ability to capture higher-order dynamics, particularly in the curvature of the yield curve, we introduce four key financial spread variables: TERMPP, FUTSWSPR, FORSPR and LIQSPRD. These variables are treated as exogenous, entering with loadings in a matrix ΦXt, adding information which debt traders are hypothesised to price in, reducing omitted-variable bias in estimates.

The TERMPP is defined as the 10-year AGS yield minus the average expected cash-rate path over the same horizon (OIS-implied), isolating the time-varying compensation for duration and macro uncertainty embedded in long-end yields. This factor primarily moves the level but also interacts with curvature during policy and balance-sheet shocks and is standard in ACM/Kim-Wright decompositions and its Australian implementation (Kopp, Emanuel and Williams, 2018). The FUTSWSPR proxies hedging pressure and balance-sheet costs that widen in macroeconomic uncertainty; these premia distort the belly of the curve and generate hump-shaped (curvature) moves that macro variables often miss. Evidence from swap/basis decompositions attributes a large share of such spreads to Treasury convenience/liquidity effects rather than pure default risk (Ying, Sheen, Gu, & Wang, 2025). The FORSPR proxies the long-horizon inflation channel with a breakeven inflation spread computed as Nominal Government yield − TIPS yield (3 years), expected to lift the level and, when long-horizon inflation premia rise more than the belly, reduce curvature. Lastly, the LIQSPRD is defined as the BBSW (3 month)-OIS (3 month) spread, a widely accepted proxy for short-term funding stress and liquidity conditions in Australian money markets. When this spread widens, it typically signals risk aversion and reduced market-making capacity, which can create kinks in the yield curve by disproportionately affecting certain maturities (Ying, Sheen, Gu, & Wang, 2025).

5.6 Model endogenous and exogenous variable selection

Strictly endogenous:

$F_t = [LOIL_t, DGDP_t, PROD_t]$

$LOIL_t = log\ of\ brent\ crude\ oil\ price\ (from\ factset)$

$DGDP_t = proportion\ of\ outstanding\ AGB\ debt\ (from\ AOFM)\ relative\ to\ aggregate\ GDP\ (from\ ABS)$

$PROD_t = monthly\ proxy\ (from\ ABS)$

Exogenous

Start with screen of Xt variables:

$X_t = [UNE_t, INFGAP_t, FUTSWSPR_t, ESBALBN_t, AUDUSD_t, TERMPP_t, FORSPR_t, LIQSPRD_t, IOREP_t]$

Further, consider transformations for the set as:

$LESBALBN = log(Exchange\ settlement\ balances\ in\ billions\ \$AUD)$

$LAUDUSD = log(AUD/USD\ exchange\ rate)$

$LIOREP = log(Iron\ ore\ price\ \$AUD)$

From the screened set Xt, we standardised variables and ran a PCA on proxies for the 3 yield curve factors in section 5.4 (in place of Wald's tests as all except LESBALN were significant). Although interpretations are not the focus of our research, we kept variables with economically sensible signs and loadings on at least one of the first three PCs (PC1=Lt, PC2=St, PC3=Ct) (see appendix 11.1.3). We therefore aggregated the final selection decision to testing whether variables add information in the system by a ≥5% FEVD share at 1-12m for at least one factor and a ≥5% RMSE reduction vs a factors-only baseline (see appendix section 11.3.1).

The final system is specified under:

$$Y_t = [L_t, S_t, C_t]$$
$$F_t = [LOIL_t, DGDP_t, PROD_t]$$
$$X_t = [UNE_t, INFGAP_t, FUTSWSPR_t, LAUDUSD_t, TERMPP_t, LIQSPRD_t]$$
$$V_t = [Y_t, F_t, X_t]'$$

Variable	Description	Data source
L	NS level factor	Factor estimated from RBA data
S	NS slope factor	Factor estimated from RBA data
C	NS curvature factor	Factor estimated from RBA data
LOIL	Log of brent crude oil price	Factset
DGDP	Growth in proportion of outstanding AGB debt relative to aggregate GDP	AOFM and ABS
PROD	Proxy for productivity	ABS
UNE	Unemployment rate AUS	ABS
INFGAP	Empirical inflation gap from expected inflation AUS	Board of Governors of the Federal Reserve System
FUTSWSPR	10 year OIS Interest Rate Swap – 10 year AGB Yield (spread)	Factset
LAUDUSD	Log of AUD/USD exchange rate	Factset
TERMPP	10 year AGS yield - average expected future cash rate (risk neutral, no arb.)	AOFM and RBA (seasonal breaks)
LIQSPRD	BBSW(3 month)-OIS(3 month)	Factset

Final system data sources (as well as the CSHR, TMINF and UTI variables used in our NS extraction) are in section 10.2. Note that the Yt and Ft blocks are treated as endogenous in estimation spaces, and the Xt set enters the system as the exogenous block.

6.0 Empirical Testing

6.1 Justification of specification

In modelling the dynamics of the yield curve, it is widely observed that the Nelson-Siegel factors, level, slope, and curvature, exhibit near-unit root behaviour (Bernanke, Boivin, & Eliasz, 2005; Barber & Copper, 2012), rendering them non-stationary in levels but stationary in first differences. However, economic theory and empirical evidence suggest that these factors are not free to drift arbitrarily but rather share long-run equilibrium relationships anchored by expectations of inflation, monetary policy, and term premia. This motivates the use of a Vector Error Correction Model (VECM), which explicitly combines short-run dynamics in the differenced series with long-run cointegrating relations among the levels. As ADF testing results (see appendix section 11.1.5) further indicate that the; DGDP, LOILP and PROD variables are I(1), and the Johansen trace test (see appendix section 11.2.3) indicates a cointegration relation of at least order r=2, the VECM specification allows as to take the first differences of I(1) variables, transforming them into I(0). Indeed, by the Granger Representation Theorem, any set of I(1) variables that share one or more long-run equilibria must specify a Vector Error-Correction representation (see also section 11.2.4).

The VECM specification provides two advantages. First, it preserves the stochastic trends inherent in yield data while ensuring that deviations from the long-run equilibrium are corrected over time. It also avoids conflating measurement noise from the yield curve with structural macro shocks, letting the state-space estimation focus on factor extraction and the VECM focus on equilibrium correction and impulse transmission. Second, it offers a framework to capture persistence, mean-reversion, and shock transmission across the yield curve in a single system. Empirical studies confirm this rationale, with Diebold and Li (2006) showing that level, slope and curvature are highly persistent and often cointegrated; Hördahl**,** Tristani and Vestin (2006) demonstrating that a VECM for yield factors aligns with macro-finance linkages and Christensen, Diebold and Rudebusch (2013) formalising this approach in the arbitrage-free Nelson-Siegel framework. This approach therefore balances statistical requirements with theoretical consistency, with stationarity tests indicating determinants are stationary (see appendix section 11.1.6). Through AIC and BIC testing (see appendix section 11.2.1), we express the VECM as a lag order of 1 (VAR lag order 2 (VAR(2)), with a restricted intercept in the cointegrating equation and no trend:

$$\Delta V_t = a_1\left(\beta_1' V_{t-1} + \rho_1\right) + a_2\left(\beta_2' V_{t-1} + \rho_2\right) + \Gamma_1 \Delta V_{t-1} + \varepsilon_t,\ with\ \varepsilon_t \sim (0, \Sigma)$$

$$a\left(\beta' V_{t-i} + \rho_i\right) =$$

$$\beta' V_{t-i} = cointegration\ vectors\ between\ variables\ (identified\ as\ S_t\ and\ C_t)$$

$$\rho_i = restricted\ intercept\ for\ non\ zero\ mean\ without\ trend\ (case\ 2)$$

$$\alpha = matrix\ of\ adjustment\ (loading)\ coefficients$$

$$\Gamma_i \Delta V_{t-i} =$$

$$Lagged\ effects\ of\ past\ changes\ on\ current\ changes,\ accounts\ for\ momentum$$

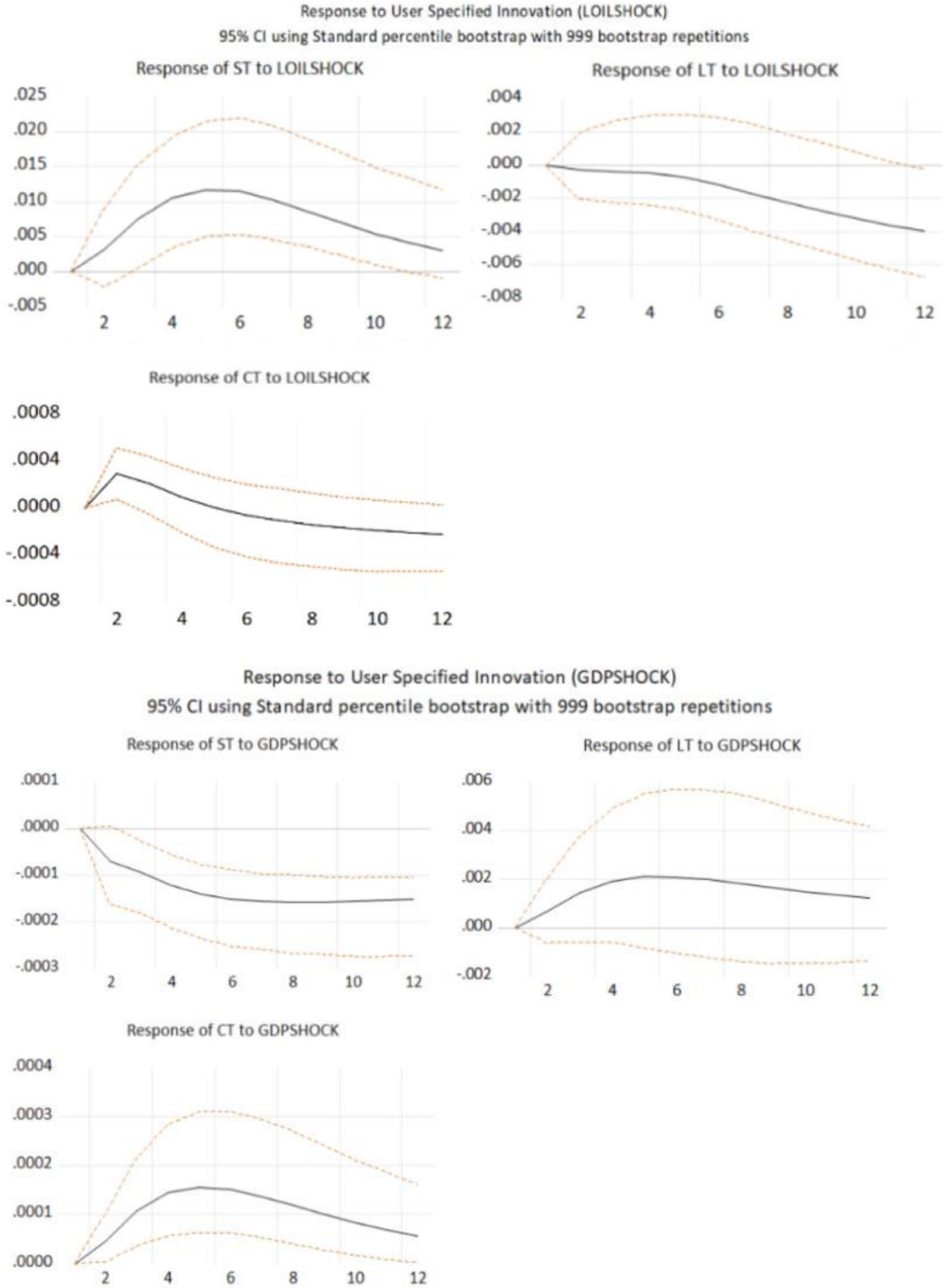
$$\varepsilon_t = residual\ vector\ which\ follows\ white\ noise\ process\ (E(\varepsilon_t) = 0)\ with\ covar\ matrix\ \Sigma$$

The estimation output for this model can be found in the appendix section 11.1.4.

6.2 Impulse response functions

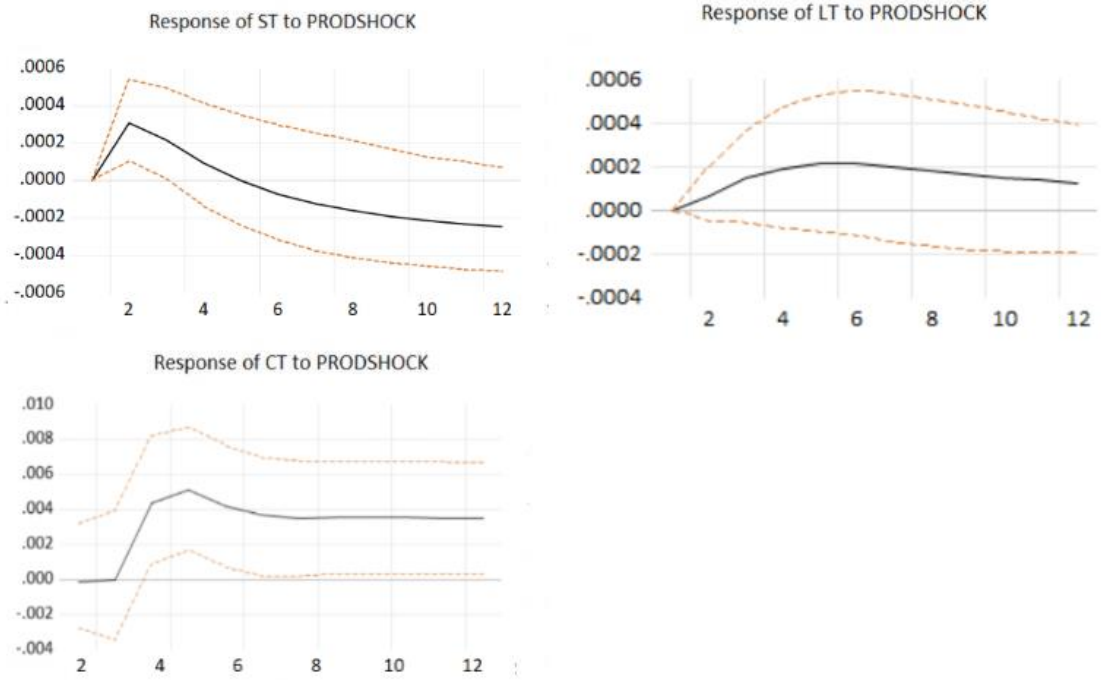
Due to the presence of autocorrelation at lags past 11 months, although the system remains stable, for a 12-month forecast our uncertainty inference is borderline. To compensate, we bootstrap forecast bands (instead of asymptotic SEs) for inference. Here, user defined innovations are generated according to the scenarios outlined in our background (section 3), with IRF responses being selected to accumulate in Eviews under a VAR specification to analyse the level effects of shocks, rather than differences in our VECM specification. For brevity, as this process is required for forecasting, see section 6.3.1 for the

specification of shock vectors to LOIL, DGDP and PROD. Note that here, unlike section 6.3.1, shocks are assumed to enter contemporaneously (at t=0), rather than being conditional at t=12.



Response to User Specified Innovation (PRODSHOCK)

95% CI using Standard percentile bootstrap with 999 bootstrap repetitions



6.3 Yield curve scenario forecast

6.3.1 Conditioning on background (section 3) meta-analysis values

As our scenario values in section 3 are predicated on 1 year ahead expectations, we treat each scenario s [for s=1,2,3] as an unanticipated conditional shock at T+12 and dynamically forecast a path from T+1, which requires saving the following data-shock vectors in EViews (applied on $\widehat{\Delta V}_{T+12}$). Adapted from section 3 findings, this requires that the variable values in T+12 months be: oil at $49USD/barrel (below 50), DGDP to increase by 2.5%YoY, and productivity to decline by 2%YoY:

$$S^{(1)}(\delta_{T+12}{}^{LOILP}) = [0, 0, 0, In(49/64.39), 0, 0]'$$

$$S^{(2)}(\delta_{T+12}{}^{DGDP}) = [0, 0, 0, 0, 0.025, 0]'$$

$$S^{(3)}(\delta_{T+12}{}^{PROD}) = [0, 0, 0, 0, 0, -\ 0.02]'$$

We first generate the unconditional (no-shock) forecast:

$$\widehat{\Delta V}^{s}_{T+1} = \widehat{\Delta V}_{T+1|T} + \varepsilon_t$$

To compute the innovation (u^*_{T+12}) required to meet the 3 targeted scenario values, we follow Waggoner and Zha's (1999) minimum distance projection method. This improves upon a standard shock vector ($\delta_{T+12}{}^{s}$) by preserving cross-equation residual structure, as well as IRFs by allowing us to simulate targeted future outcomes rather than generic one-time shocks, ensuring the system evolves along a path consistent with specified policy scenarios.

$$u^*_{T+12} = \Sigma H'(H\Sigma H')^{-1}(\delta_{T+12}{}^{s} - H\widehat{\Delta V}_{T+12|T}),$$

Σ= variance-covariance matrix of reduced-form forecast errors (i.e. residuals) from the VECM.

H= selector matrix for conditioning on LOILP, DGDP and PROD where $H \in R^{3\times n}$

We then recursively forecast under scenario s, using the VECM dynamic equation for $h \geq 2$ [for h=2,3….12]:

$$\widehat{\Delta V}^{s}_{T+h|T} = a(\beta'\widehat{V}^{s}_{T+h-1|T} + \rho) + \Gamma_1\widehat{\Delta V}^{s}_{T+h-1|T} + u^*_{T+h}$$

$$\widehat{V}^{s}_{T+h|T} = \widehat{V}^{s}_{T+h-1|T} + \widehat{\Delta V}^{s}_{T+h|T} \textit{ (levels result)}$$

Here, it is important to note that we are therefore not shocking via innovation identification, but instead, conditioning the data directly to *T*+12 in line with background research. Therefore, an alternative framework to consider, like our IRF methodology, could instead be to assume these shocks enter contemporaneously (at t=0) and model to T+12. Here, the VECM dynamics propagate these conditional paths to L, S, and C depending on how strong the system-wide linkages are with the other variables (as captured by α and Γ's).

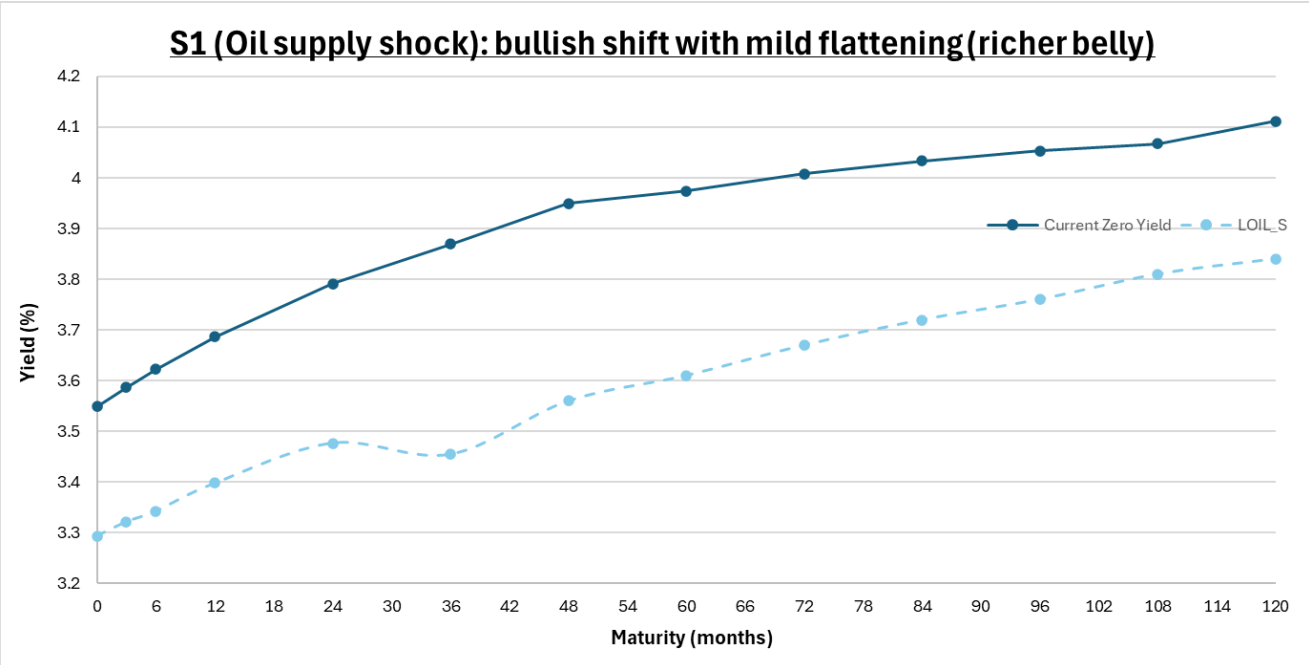
6.3.2 Discussions on this approach

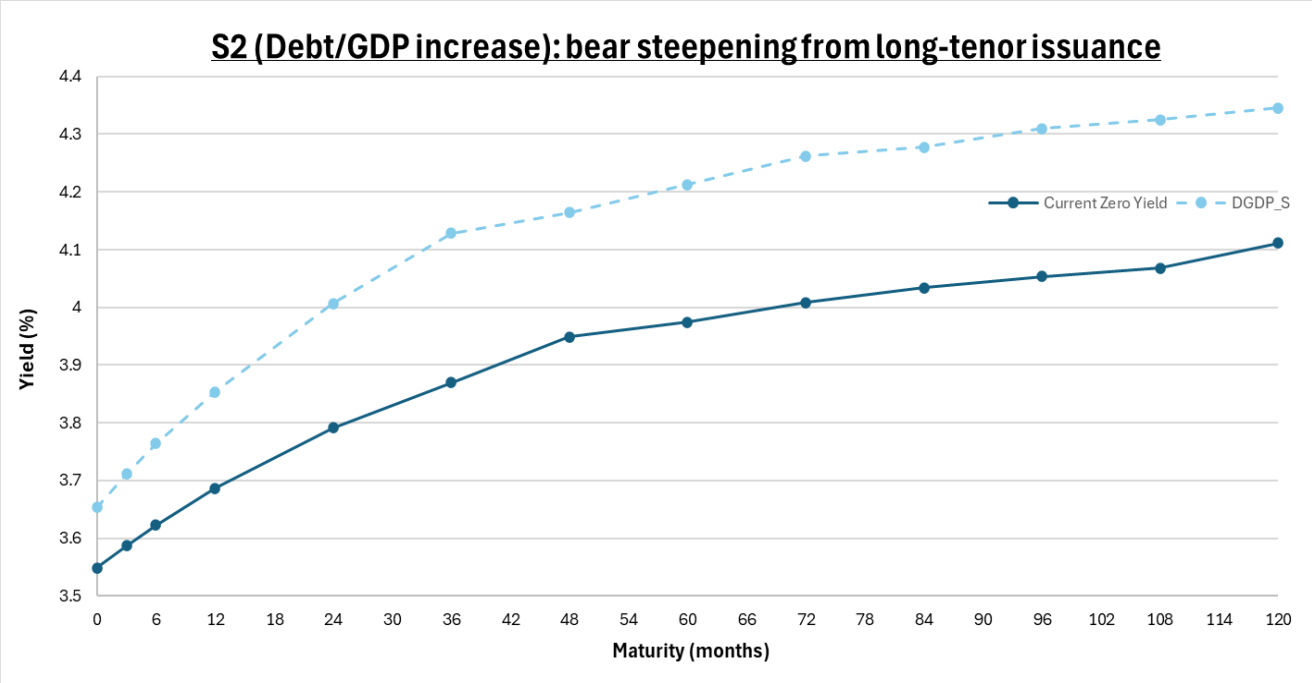
Although we apply shocks only to three endogenous variables, this framework (sections 5-6) is fully extensible, enabling any user-defined scenario to be incorporated, whether based on fiscal projections,

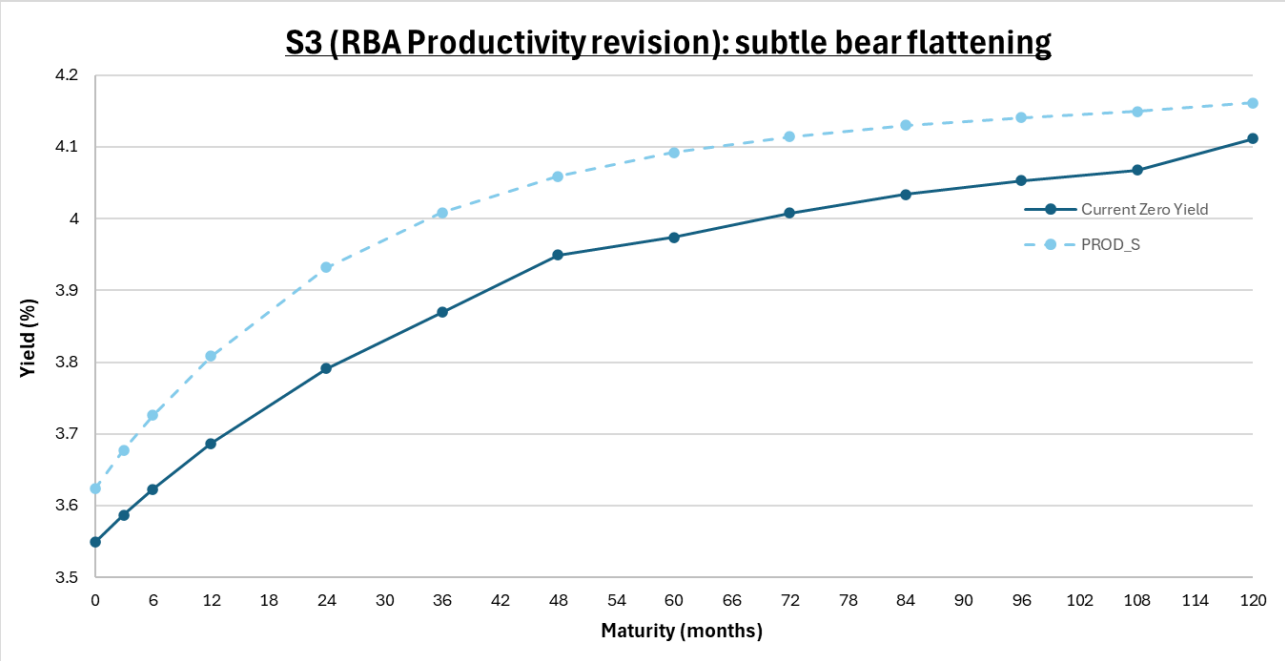
commodity price assumptions, or (as in our case) meta-analysis, making it a practical tool for policymakers and debt portfolio managers alike. While our scenario shocks are clearly defined, a valuable extension would be to model their likelihood using historical data to construct probability distributions. This would allow for richer risk-cognisant interpretations, particularly relevant for policy and portfolio applications. However, due to constraints in data availability and time, we relied on deterministic shocks rather than simulating full probability distributions. Future work could address this by integrating regime-switching models or stochastic simulation techniques to better quantify the uncertainty of the shocks we defined.

6.3.3 Fitting forecasted system into the yield curve scenarios

To visualise the yield curve implications of each scenario, we extract the forecasted Nelson-Siegel factor paths F=[Lt,St,Ct] generated by the VECM(1) model at T+12. As the VECM(1) generates forecasts for factors as differences, i.e. ΔF_{T+1}, to represent the level of a factor at each forecast horizon, Eviews provides adjusted output as the cumulative sum (integration) of forecasted differences: $F_{t+h} = F_t + \sum_{j=1}^{h} \Delta F_{T+j}$. These factor levels are then inputted into the Nelson-Siegel specification (section 5.1) to compute the term structure of yields across maturities τ={0,…,120}. Section 11.3.2 of the appendix includes forecast diagnostic results. Note that for τ=0, the expected (current) cash rate is equivalent to the level term + the error measurement due to NS formula division errors for a τ=0 value. This limitation is shown to provide different 0month-12month premium dynamics across graphical results (see also section 11.3.2).







6.4 Modelling considerations

Our decision to model the yield curve using a dynamic Nelson Siegel framework with a VECM macro and fiscal variable block reflects a trade-off between theoretical rigor and empirical performance, as the model does not allow for clear decomposition of yields into expectations and term premia like other studies (see Cohen et al., 2018). Indeed, we adopt a semi-atheoretical, reduced-form approach that intentionally avoids the tight no-arbitrage constraints typically imposed in affine term structure models (see Dewachter et al., 2013 p.5 for a useful extensible example). This prioritizes tractability and forecasting ability over theoretical consistency, particularly useful in our data-limited and undergraduate context. However, for the purposes outlined in sections 2 and 8, this design choice is supported by extensive literature, where Diebold and Li (2006) show that the three-factor Nelson-Siegel model delivers competitive out-of-sample forecasts relative to more complex arbitrage-free models, where Duffee (2002) further finds that imposing no-arbitrage restrictions across maturities rarely

improves forecast accuracy. Joslin, Singleton and Zhu (2011) also demonstrate that the conditional forecasts of yield curve factors are largely invariant to whether those restrictions are imposed.

We also confront a trade-off between model complexity and interpretability, where we favour the original three-factor Nelson-Siegel specification for its parsimony and clarity, as each factor (level, slope, and curvature), can map directly to an intuitive segment of the yield curve for our communication of results. More complex representations, such as the Svensson (NSS) extension, introduce a second curvature term and decay parameter to capture additional humps, which may better fit Australian yield curves that often display a pronounced belly around 3-7 years and, at times, a second kink further out due to changes in issuance policy or long-end demand. While NSS is widely used by central banks for this reason (Svensson, 1994), the additional parameters introduce potential estimation challenges, including higher sampling variance and weak identification of the second decay term in short or noisy samples. To be candid, this is also a rather practical issue, as we had difficulties learning and implementing the state-space transition and measurement equation framework in our undergraduate context (much of the estimation time was spent on this). These issues also complicate factor interpretation and increase the risk of overfitting. Hence, we preserve the simpler NS form for baseline modelling and recommend NSS as a potential extension only if it demonstrates meaningful out-of-sample improvement. However, these limitations require us to remain cautious when drawing economic inferences from the estimated factors.

In the estimation of our historical NS-SS yield factors, we allowed the NS decay parameter (λ) to be estimated freely each period, as Annaert et al's. (2010) grid search evidence indicates that the optimal parameter moves substantially over time, with the fixing of the parameter creating specification risk that reduces comparability amongst data sets and introduces multicollinearity between the slope and curvature factors. However, to stabilise forecast estimation requirements, we fix the NS decay parameter (λ) to a value that positions the curvature peak in a policy-relevant maturity range in section 6.3 for all yield curve scenarios. This practice is common but not without trade-offs. Annaert et al. (2010) show that fixed λ settings can introduce bias if curve shapes evolve materially, particularly when issuance structure or demand shocks shift the belly of the curve. Beyond the curve specification itself, richer macro-financial variable sets, such as forward-rate-based "tent" factors (Cochrane & Piazzesi, 2005), maturity-weighted bond supply, or risk aversion proxies like the VIX, are shown to enhance forecasts. However, incorporating such predictors into a standard VAR risks overfitting. A Bayesian VAR (BVAR) with Minnesota-type shrinkage would offer a tractable solution, improving stability and forecast performance even in high-dimensional settings (Bańbura, Giannone & Reichlin, 2010), although beyond our research scope as undergraduates. Together, these considerations offer a well-defined roadmap for refining the model.

7.0 Results and Discussion

Results from the yield curve forecasts are presented below, followed by a discussion of their implications. Each scenario illustrates how macroeconomic shocks transmit through financial markets to influence the shape and dynamics of the yield curve.

7.1 Scenario analysis

7.1.1 Oil Price Decline Scenario

Our model concludes that an oil price decline induced from OPEC overproduction will result in a bull steepening trend, characterised by a fall in the level component, an increase in the slope factor, and a more negative curvature to create a richened belly. Anticipating this term structure (i) assists households in choosing cost-effective borrowing structures, (ii) and elucidates optimal bond holdings for investors to maximise returns in a disinflationary environment.

Level

World Bank (2024) shows that during changes to OPEC production scale, the standard deviation of daily log oil price changes momentarily exceeds 4.6% before receding to its long-run average of ~2.4%, underscoring its historically high and short-lived volatility. Resultantly, our model assumes the decline to US$50/bbl to be a transitory impulse which will decay over time, with prices reverting to a mean of low-US$60s over late 2026. This can be illustrated by the distance between our modelled yield and the current yield curve being more pronounced at the 1–3-year maturity bucket (dropping by 30bp on average between 1-3years), and narrowing for higher order maturities (declining by 24bp on average between 8-10years), implying the oil price shock is most potent over the short term. Moreover, the RBA (2025) empirically concludes that a 22% fall in oil prices from current levels of $64.39 to $50, will reduce headline inflation by approximately 50–60 bps as firms pass on cheaper production costs to consumers in order to preserve competitiveness without compromising profit margins. With June quarterly CPI at 2.1%, the transmission of lower oil prices will cause inflation to dip below the 2-3% target band, necessitating expansionary monetary policy. Supportingly, future cash rate cuts over 2025/26 are echoed by the big 4, akin to our 0-year bond yield forecast (which can be considered a proxy for the cash rate) of 3.29% when interpreted through the Nelson Siegel approach. Effectively, a lower cash rate reduces the return overnight lending, increasing the opportunity cost of higher yielding government securities. As institutions rotate into government bonds, surging demand drives up prices and suppresses shorter term yields, while longer run rates remain anchored as oil prices normalise.

Bank	***2025***	***2026***
Westpac	25bp cut in November to take the cash rate to 3.35% by year-end.	Further easing through 2026 to as low as 2.85%
Commonwealth Bank	Rate cut in November to 3.35%	N/A
National Australia Bank	N/A	Rates drifting lower to 3.0%-3.1% throughout early 2026.
ANZ	3.35% by November-end	N/A

Meaning for to Households

Given the level of the curve is likely to drop materially below today's levels, households and businesses should favour fixed-rate borrowings at the commencement of the oil-induced easing cycle. As aforementioned, an OPEC-driven oil oversupply is inherently transitory, so its disinflationary impulse and the rate cut it triggers can reverse, leaving longer term variable-rate borrowers exposed if monetary policy normalises. Specifically, APRA (2025) quantifies a 25bp cash rate reduction to lower interest costs by roughly A$130 per month per mortgaged household. Applying this empirical rationale to our

anticipated 0m bond yield of 3.3% – i.e. ~30bp decline from the current cash rate of 3.6% – opting for a fixed rate loan in this environment will raise household discretionary incomes by roughly $156 per month over the life of the loan.

Meaning for to Investors

Moreover, with current bond yields overstated ahead of the imminent level drop, investors should usurp these relatively cheap securities and offload their position once monetary easing commences. The cash rate decline will create a surge in demand for previously issued off-the-run higher yielding bonds, bidding up price levels and lowering yields. The table below confirms a downward shift in the yield curve to induce price gains at all bond maturities.

Duration pricing formula for a level yield change: $\frac{P(i_1) - P(i_0)}{P(i_0)} \approx -(i_1 - i_0) \times MD$

Where subscript '*0*' is the current realised yield curve, and subscript '*1*' is our forecasted yield.

Maturity (Months)	Maturity (Yrs)	Duration (Yrs)	MD (Yrs)	Current Yield (%)	Oil Shock Yield (%)	Change in level (%)	Price change (%)
0	0	0.00	0.00	3.55%	3.29%	-0.26%	**0.00%**
3	0.25	0.25	0.24	3.59%	3.32%	-0.27%	**0.06%**
6	0.5	0.50	0.48	3.62%	3.34%	-0.28%	**0.13%**
12	1	1.00	0.96	3.69%	3.40%	-0.29%	**0.28%**
24	2	2.00	1.93	3.79%	3.48%	-0.31%	**0.60%**
36	3	3.00	2.89	3.87%	3.46%	-0.41%	**1.20%**
48	4	4.00	3.85	3.95%	3.56%	-0.39%	**1.50%**
60	5	5.00	4.81	3.97%	3.61%	-0.36%	**1.75%**
72	6	6.00	5.77	4.01%	3.67%	-0.34%	**1.95%**
84	7	7.00	6.73	4.03%	3.72%	-0.31%	**2.11%**
96	8	8.00	7.69	4.05%	3.76%	-0.29%	**2.25%**
108	9	9.00	8.65	4.07%	3.81%	-0.26%	**2.23%**
120	10	10.00	9.61	4.11%	3.95%	-0.16%	**1.55%**

Slope

The slope of the yield curve is poised to steepen as near-term easing weighs down the short end, while the long end remains partly anchored, substantiating the decayed explanatory power of oil prices over time. Long run yields are dominated by structural factors and term premium, remaining largely insensitive to temporary oil price shocks. This can be reinstated by the IMF: *"Expectations about structural factors like inflation or potential growth are revised only slowly, even after large shocks, pointing to high and persistent inertia"* (Macr--osynergy Ltd, 2018). Moreover, with the IEA (2025) concluding that 70% of Australia's oil consumption is tied to transportation, the Australia Treasury integrates growing electrification through rising EV uptake (projected to have a 33.24% CAGR over 2025-2030) and renewable energy transition assumptions in their macro and sectoral frameworks. Specifically, the Treasury September 2025 release *"Australia's Net Zero Transformation" mentions that for all forecasted economic data, they factor in "strong investment in renewable energy…enabling broad-based decarbonisation through electrification [&] fuel switching"*. By using this Treasury data as exogenous variables for our projections, our model implicitly incorporates the underlying structural trend of an oil-phaseout, with this commodity bearing a muted impact on longer run inflation and hence yields.

Meaning for Households

A steeper forecast yield curve lifts the 10y–3m spread - a common proxy for banks' net interest margins - from 52bp to 63bp. Banks perform maturity transformation: they price long-dated loans off long-term rates while funding with short-term deposits, and when this spread widens and margins improve, lenders are more willing to write mortgages and business loans, easing refinancing and credit access for households. Recently, the 10y–3m spread reached a 2-year peak of 0.42 in January 2025, coinciding with a 0.3% qoq rise in lending – which is anticipated to move higher in this oil oversupply scenario (FRED, 2025). Correspondingly, a steeper yield curve is often associated with a more favourable lending environment climate, defined by higher new loan issuances, stronger intra-bank competition, and compressed net interest margins that translate into improved borrowing conditions.

Meaning for Investors
Moreover, stability to the far end of the yield curve is ideal for low-risk vehicles, namely pension funds which hold onto long-dated securities to appeal to risk averse retirees biased to capital preservation. This is echoed by the Monash University survey (2018) which concludes that pension funds, on average, allocate 61.26% of their total assets to defensive securities, namely fixed income. Therefore, locking in long-dated fixed income securities effectively hedges against oil-induced price volatility, safeguarding risk-off returns. Such behaviour can be seen in the above table, with the rate of price change declining from 2.25% in the 8-year bucket, to 2.23% and 1.55% in the 9- and 10-year buckets respectively, signifying growing inflexibility and price stability toward the far end of the curve relative to the mid-segment, deeming it a suitable risk off investment.

Curvature
Yield curve steepening is often accompanied by a richening of the belly, as investors gravitate to medium-term securities which provide the most favourable risk and return trade off, measured through the carry-rolldown per unit of duration. As aforementioned, short-term yields decline immediately once markets price in imminent monetary easing, while longer-dated yields remain anchored by term premiums and structural influences, meaning the 2–4-year segment captures the steepest localised slope – giving the highest drop in expected short rates (Miyazaki, et al, 2021). Specifically, the carry refers to the return you earn if the curve level and shape remain fixed over the horizon (i.e. its current coupon rate), whereas the rolldown is the capital appreciation incurred when the bond "slides" to a shorter, lower yield point on the curve. Also, investing at this segment mitigates price risk common with long duration bonds – which are highly sensitive to interest rate changes since coupons will be discounted over a longer maturity. DBS Group Research (2016) reports a similar phenomenon during the oil price collapse on the 2010-2012 UST curve, noting "oil prices continuing to drift lower and wages showing no sign of acceleration", creating a bull steepening trend with immediate monetary loosening and "modest" changes to longer-term yields. As markets priced in near-term easing, but shallow rate cut expectations over the long run, the belly offers the strongest rolldown capital gains, prompting investors to concentrate in mid segment USTs. Continued investment in the mid segment compressed yields, enhancing rolldown gains further, and creating a feedback loop which reinforced even higher demand, pushing belly yields lower and accentuating the bowing shape (Leow, E. Kert, J. 2016). Overall, we expect a similar dynamic to occur in the Australian economy, as demand concentrated to the intermediate sector – where the risk return profile is most optimal – should richen the belly of the curve.

Meaning for Investors
Ultimately, roughly 60% of Australia's tradeable government bond floats are held by profit-seeking investors, meaning a wide cross-section of the market can benefit from carry-rolldown opportunities. To maximise return per unit of risk, tilt holdings to the 4-year bucket and hold for an additional year (selloff at year 3), offering the highest capital gains potential since it possesses the steepest localised slope. As mentioned in the table below, approximately every 10 bps of 4y to 3y roll-down is worth ~0.40% in price appreciation, with a carry of ~0.11% on a typical 4-year note; generalising to a peak of 13bps return per unit of duration— meaning a 4-year hold-for-one-year trade often maximises capital gains per unit of return.

$$Carry = (y_{future} - y_{current}) \times (MaturityYrs_{future} - MaturityYrs_{current})$$
$$Rolldown = (y_{future} - y_{current}) * MD$$
$$Carry + Rolldown\ per\ unit\ of\ duration = \frac{Carry+Rolldown}{Duration}$$

Maturity (Months)	Maturity (Yrs)	Duration (Yrs)	MD (Yrs)	Oil Shock Yield (%)	Carry (%)	Rolldown Price change (%)	Carry Rolldown per unit of Duration
0m	0	0.00	0.00	3.29%			
3m to 0m	0.25	0.25	0.24	3.32%	0.01%	0.01%	0.06%
6m to 3m	0.5	0.50	0.48	3.34%	0.01%	0.01%	0.03%
12m to 6m	1	1.00	0.96	3.40%	0.03%	0.05%	0.08%
24m to 12m	2	2.00	1.93	3.48%	0.08%	0.15%	0.12%
36m to 24m	3	3.00	2.89	3.46%	-0.02%	-0.06%	-0.03%
48m to 36m	4	4.00	3.85	3.56%	0.11%	0.40%	0.13%
60m to 48m	5	5.00	4.81	3.61%	0.05%	0.24%	0.06%
72m to 60m	6	6.00	5.77	3.67%	0.06%	0.35%	0.07%
84m to 72m	7	7.00	6.73	3.72%	0.05%	0.34%	0.06%
96m to 84m	8	8.00	7.69	3.76%	0.04%	0.31%	0.04%
108m to 96m	9	9.00	8.65	3.81%	0.05%	0.43%	0.05%
120m to 108m	10	10.00	9.61	3.95%			

7.1.2 Debt-to-GDP scenario

An upside shock to the debt-to-GDP ratio generates a bear steepening trend for the yield curve, represented by longer-dated yields increasing more than short-dated yields. Translating this into the NS components, we observe a steepening of the slope factor (Δ 0.0013) over the forecast horizon. In contrast, the curvature factor (Δ 0.0017) shifts outward, and the level (Δ 0.0019) shifts upward. The increase in the level reflects an upward repricing of yields consistent with higher expected inflation and fiscal risk. The steepening of the slope indicates a divergence between short-term monetary policy expectations and long-term bond issuance levels. The increased curvature suggests that investors require greater compensation for holding medium-term bonds, emblematic of growing fiscal concerns around the middle of the curve.

IRF results show that a one standard deviation change to the debt-to-GDP ratio is equivalent to a 2.50% increase to the ratio, which equates to a 24-basis-point increase in the 10-year yield. This is supported by Poghosyan (2013), who found that a one-per-cent increase in government debt-to-GDP ratios in advanced economies leads to a 2-basis point increase in 10-year government bond yields. The forecast accuracy generates an RMSE of 0.087 and an adjusted R-squared of 0.215 across the 12-month horizon. Movements are most profound for maturities beyond 2 years, suggesting that fiscal dynamics mostly impact the middle to back end of the yield curve, which is broadly consistent with Afonso & Martins (2010).

This can be interpreted as the yield curve being influenced through the expectations and term-premium channels. Investors anticipate future monetary policy tightening in the presence of inflationary pressure as debt levels rise, while requiring greater compensation for duration risk due to higher fiscal uncertainty. Our results are persistent over the 12-month forecast horizon, with only the curvature showing signs of reversion after nine months, which implies there is minimal mean reversion in yields following fiscal deterioration. Following Challenger (2025), worsening budget sustainability amplifies the effects on the term premium, exerting upward pressure on long-term yields.

Meaning for Households

The effects of the bear steepening will be predominantly transmitted to households through an upwards revision of long-term mortgage rates. Fixed mortgages are relatively priced using AGB yields with corresponding maturities as the 'risk free rate', in addition to premiums accounting for credit risk and operational margins. Banks adjust mortgage rates in line with movements in this risk-free term structure, hedging consumer exposures using bonds and interest rate swaps (Ambrose and Lacour-Little, 2001). As the fiscal spending pressures cause longer-term yields to rise, so too do fixed mortgage rates. This dynamic was clearly observable between 2021-22, when the 3-year AGB yield rose by +242 basis points

and coincided with a 1-2% increase in advertised 3-year fixed mortgage rates (*based off data from https://db.nomics.world/RBA/F2.1*).

In this environment, shorter maturity loans or adjustable-rate mortgages (ARMs) become comparatively more attractive, as their rates are anchored closer to monetary policy paths and therefore experience weaker shock loadings (Campbell and Cocco, 2003). Shifting towards shorter reset horizons increases exposure to refinancing risk and future policy uncertainties, since households face the possibility of repricing into higher short-term rate environments. The decision to shift towards adjustable rates should therefore vary between households, depending on risk tolerance and income restraints.

Meaning for Investors

For portfolio managers, a rise in term premia at the long end of the curve reduces the valuation of duration-sensitive assets. This encourages a rotation into shorter-maturity securities and floating rate instruments that carry lower duration exposures to AGBs (Greenwood, Hansen, Rudolph and Summers, 2015). With long-end yields rising because of perceived fiscal risk, government bonds become volatile and tend to behave like riskier securities. The historically negative correlation between stocks and bonds edges higher towards zero, thus weakening the hedging value of government bonds. This occurred during the high inflation periods of the 1970s and during the European sovereign debt crisis of 2011 (Koijen, Koulischer, Nguyen and Yogo, 2017).

With curvature also rising, the required compensation for mid-term maturities becomes higher. Roll down strategy becomes less reliable as forward rates are based more on term-premium dynamics rather than market expectations alone, making it harder to predict future rate paths. As a result, balanced investors aim to minimise portfolio duration and pivot towards inflation-linked bonds.

7.1.3 Productivity Scenario

A downward revision to productivity has induced a mild bear flattening trend in the yield curve, shown by a relatively even increase in yields for both short and long-dated maturities. This appears inconsistent with the standard relationship, in which lower productivity reduces potential growth and, hence, neutral real interest rates. In contrast to our results, Holson, Laubach & Williams (2017) found that neutral real interest rates decline as future GDP growth expectations decline. The temporary rise in yields in our model may instead reflect transitory inflation risk or term-premium adjustments, rather than a structural increase in neutral real interest rates.

A one standard deviation change is equivalent to a 2% downward revision to productivity YoY, which has the largest impact on the 2 and 3-year yields, increasing by 14.6 and 13.8 basis points, respectively. The forecast accuracy generates an RMSE of 0.0746 and an adjusted R-squared of 0.305 across the 12-month horizon. This pattern contradicts what was found by Diebold, Rudebusch, and Aruoba (2006), who measured the yield curve's response to a positive shock to potential output led to a mild bear flattening.

Decomposing this using the IRF framework, we see an initial flattening in the slope ($\Delta 0.0035$) that reverts to its baseline after six months as short-term interest rates adjust to revised expectations. A persistent outward shift in the curvature (Δ 0.00316) that peaks at five months, implying that medium-term yields rise disproportionately to the front and back end of the curve. The level factor has a persistent upward shift (Δ 0.001718) over the forecast horizon, which suggests a sustained upward repricing of yields.

These results suggest that Australian yield curve responses to a downgrade in productivity are influenced by broader macro-financial factors. The upward shift in yields may indicate markets'

perception of fiscal-sustainability risks or heightened inflation. This may signal to investors a deterioration in supply-side fundamentals, warranting a higher risk premium and constraining the central bank's scope for monetary easing. Rather than signalling optimism, the upward repricing reflects a precautionary response, where structural headwinds to growth are priced into bond markets through higher long-term rates. Overall, the results illustrate how adverse productivity dynamics can yield counterintuitive increases in yields, driven by expectations of reduced macroeconomic resilience and greater uncertainty around future real interest rate paths.

Meaning for Households

The upwards shift in yields can be interpreted as a rise in term premia associated with heightened uncertainty, rather than evidence of a structural change in long-run policy implications. For households, the positive change in 2–3-year yields, the largest on the forecasted yield curve, would cause a reduction in refinancing incentives, as rolling into new loans at maturity would now occur at a higher rate. Moreover, the relatively small adjustments at the long end of the curve, consistent with a bear flattening, make longer-maturity fixed loans more appealing, since those rates will not have risen to the same extent under our scenario.

Furthermore, the downgrade in productivity also implies weaker real wage growth, even if nominal wages were to rise in the short-term. This aligns with Mankiw (2006), who found that productivity is the fundamental long-term determinant of real income. Households are therefore dealt with higher borrowing costs (from previous paragraph) and a deterioration in expected wage growth. This combination reinforces precautionary saving behaviour and discourages household investment (Carroll, 1996).

8.0 Policy Implications

The government must work in partnership with the Reserve Bank of Australia (RBA) as well as the Australian Office of Financial Management (AOFM) ensuring an active role in reshaping the structure and timing of public debt issuance in response to yield curve dynamics produced by scenario-based impulse response functions (IRFs). These IRFs which are derived from macroeconomics shocks such as rising debt-to-DFP ratios, productivity, or oil supply disturbances indicate meaningful shifts in the yield curve that should directly inform fiscal and monetary coordination.

8.1 Strategic Extension of the Debt Maturity Profile

One way this can be achieved is through optimising the maturity mix by issuing long-term bonds. IRFs that project long-term yields, particularly under a scenario of rising debt-to-GDP, call for pre-emptive issuance of long-dated securities of 10-20 years before market rates begin to climb. This enables the government to lock in favourable, lower borrowing costs before the yield curve steepens further, protecting the fiscal position against higher interest burdens in future borrowing cycles. Empirical research supports this approach: lengthening debt maturities reduces rollover and interest rate risk while anchoring long-term debt sustainability (Ciaffi et al., 2024).

Furthermore, the AOFM has already demonstrated this approach has effectively been implemented in past cycles, most notably during the post-GFC period when it extended the Commonwealth AGB curve from 12 to 30 years and increased the weighted average term-to-maturity from roughly 6 to 10 years to managing refinancing and interest rate risk. They explain that they conducted this lengthening to reduce refinancing peaks, smooth future borrowing programs, and enhance market liquidity (Australian Office of Financial Management, n.d.). Hence replicating such strategy under current conditions would be essential, particularly as long end IRFs indicate upward pressure from larger fiscal deficits and inflationary expectations. Fundamentally, advancing long-term issuance would enhance funding certainty and preserve the government's capacity to respond flexibly to future shifts in market risk perception.

8.2 Aligning Fiscal Strategy with Monetary Policy Objectives

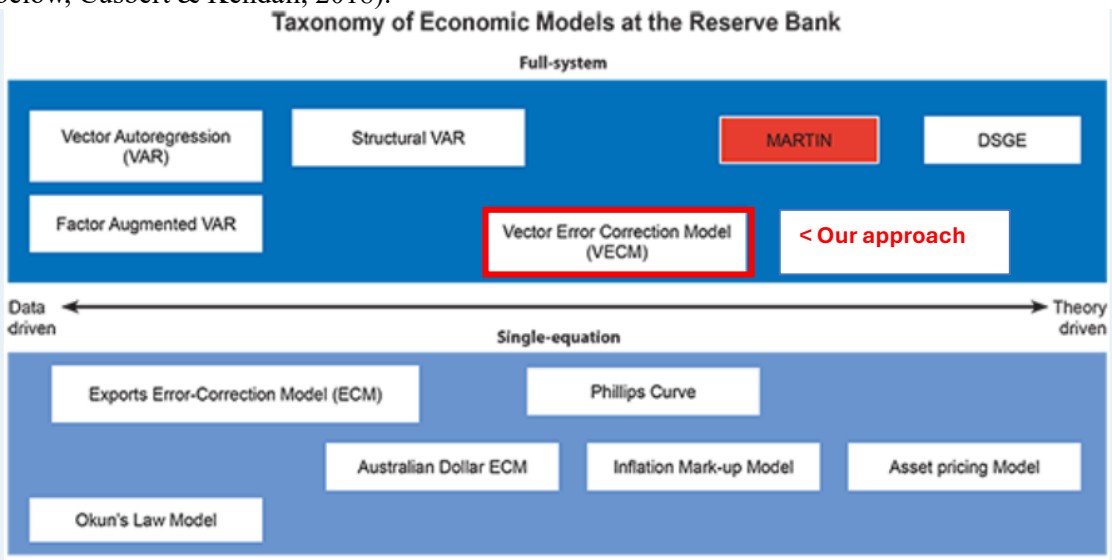
To complement maturity-based strategies, coordination between fiscal authorities (AOFM) and the RBA is essential, as liquidity policy tools move the curve independently of issuance. Empirically, during COVID-19, the Term Funding Facility (TFF) provided low-cost three-year funding, dampening bank bond issuance and compressing 2-4-year rates. Under Basel III, the LCR's and NSFR's and, until its 2022 phase-out, the Committed Liquidity Facility (CLF) also reshaped banks' structural demand for HQLA (AGS/semis) (Reserve Bank of Australia, 2024). Together with QE, these settings reduced the free float of bonds and flattened the curve, limiting the effectiveness of standalone monetary policy operations (as the AOFM has noted via FOI 3206).

Because our VECM utilises conditional variable forecasts, not policy determinants, the yield path is uncertain and sensitive to future TFF maturities, QE reinvestment, and regulatory liquidity driven changes. To mitigate volatility and avoid policy friction, we therefore recommend jointly signalled issuance calendars, coordinated forward-guidance on RBA balance-sheet and liquidity operations, with transparent reporting of free-float and HQLA demand so markets can price term premia with clearer expectations. Indeed, Banerjee (2024) expresses that when governments and central banks respond jointly to inflationary supply shocks such as oil prices, public debt ratios tend to fall and fiscal balances improve in the medium term as simultaneously occurring actions prevent an 'excessively contractionary macroeconomic stance.' In the Australian context, aligning the Treasury's debt strategy with RBA rate expectations would steady yields, reinforce investor confidence, and ensure greater consistency and predictability in government financing costs.

Consequently, fiscal and monetary synchronisation transforms our modelling projections from passive indicators into actionable guidance, ensuring that debt strategy supports rather than offsets monetary objectives.

8.3 Economic modelling for Central Banks

In a policy environment characterised by persistent shocks and evolving risks, central banks require modelling systems that are not just statistically sound, but also transparent, flexible, and communicable. Our approach, linking state-space extracted yield curve factors with macro-fiscal variables in a VECM system, strikes this balance. Like the RBA's MARTIN model, it operates within the "full-system, semi-structural" space: theory-guided, but data-driven, and specifically built to reflect long-run equilibrium while preserving short-run dynamics. This supports a more faithful representation of economic persistence and feedback than single-equation models or data-driven machine learning tools (see image below, Cusbert & Kendall, 2018).



By striking this balance, the model is structured for practical policy applications. Analysts can apply conditioning paths on other AS/AD shocks without re-estimating the entire system. This allows fast scenario iteration for monetary, fiscal, or debt strategy deliberation, supporting real-time stress testing and communication of trade-offs. Compared to rigid frameworks like no-arbitrage term structure models or DSGEs, our semi-structural design accommodates uncertainty while ensuring each modelling component remains interpretable. In doing so, we mitigate the "black box" risk flagged in the RBA's modelling taxonomy, where more opaque methods often hinder policy communication.

9.0 Conclusion

This study develops an integrated macro-finance framework to forecast Australia's government bond yield curve under major macroeconomic shocks. Using the Dynamic Nelson-Siegel model with a Vector Error Correction approach, it examines how fiscal conditions, commodity prices, and productivity shape interest rates. The results show distinct dynamics - oil price declines cause bull-steepening through monetary easing, rising debt-to-GDP leads to bear-steepening from fiscal risk and weaker productivity drives mild bear-flattening via higher neutral rates. The yield curve thus reflects and transmits macroeconomic forces. For policymakers and investors, scenario-based strategies across debt maturity, communication, and coordination enhance stability, credibility, and foresight.

10.0 References

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11.0 Appendix

11.1 Model outputs not included in main report body

11.1.1 Estimated Q matrix

Covariance Analysis: State Innovations
Date: 09/23/25 Time: 21:59
Sample: 1992M07 2025M08
Included observations: 398

Covariance	LT	ST	CT	UTI	CSHR	TMINF
LT	0.0003340					
ST	0.0000540	0.0000545				
CT	-0.0000115	-0.0000019	0.0000397			
UTI	0.0002552	0.0000309	-0.0000147	0.0021670		
CSHR	0.0000198	0.0000016	-0.0000019	0.0000181	0.0000187	
TMINF	0.0000520	0.0000239	0.0000008	0.0003609	0.0000201	0.0001670

Probability	LT	ST	CT	UTI	CSHR	TMINF
LT	-----					
ST	0.0124	-----				
CT	0.2140	0.0216	-----			
UTI	0.0739	0.2374	0.3144	-----		
CSHR	0.0480	0.1845	0.2577	0.2001	-----	
TMINF	0.0714	0.0489	0.3556	0.0120	0.0312	-----

11.1.2 Selection of loading factor

We allow lambda to be freely estimated across time for historical state space estimates, using a Kalman filter. However, for measuring the yield curve in 2025 M8 onwards, as Diebold and Li (2006) select =0.0609 for a peak hump loading at 29.5 months, and the empirical yield data indicates a hump closer to 36 months for Australia, we select =0.0498. On AU zero-coupon curves (3m–10y grid), a coarse grid-search over $\lambda \in [0.04, 0.08]$ showed the cross-section RMSE is flat around 0.04-0.06 with a shallow minimum near 0.05-0.055. The 3-year AGS is a benchmark maturity (historically the RBA's yield-curve control targeted at the 3-year point), with medium-term policy news loading most strongly around 3y, not 2.5y. AOFM issuance and outstanding stock are also concentrated in 2–5y and 10y buckets, with the 3y region often bearing the brunt of relative-value and supply shocks, which is exactly what the NS curvature factor is meant to capture.

11.1.3 Selection of exogenous variables

Eigenvectors (loadings):

Variable	PC 1	PC 2	PC 3
LT	0.748855	-0.169319	-0.103345
ST	0.031059	0.635005	-0.188727
CT	0.037937	-0.091929	0.468107
PROD	0.284010	0.283617	-0.191494
DGDP	0.134436	0.370189	-0.196930
LOILP	0.241898	-0.216055	0.098854
UNE	0.214512	-0.165831	0.066776
INFGAP	0.353161	-0.145280	-0.093444
LAUDUSD	0.254654	0.134814	0.277645
LESBALBN	0.015115	0.025491	0.026756
TERMPP	0.122696	0.203030	-0.239759
FUTSWSPR	0.052416	0.346619	0.540890
LIQSPRD	0.137618	0.223249	0.437613
FORSPR	0.083712	0.129823	0.129651

Ordinary correlations:

In place of Wald's tests (as all except LESBALN were individually significant, with LESBALN, FORSPR and LIOREP jointly insignificant), we use PCA, but only as a theory-guided pre-screen on the joint [L,S,C|X] set to check sign/association patterns, as it is purely a-theoretical. The final inclusion

decision is instead made by in-sample FEVD and RMSE contribution tests relative to a factors-only benchmark. See section 11.3.1. A useful extension of our modelling framework would involve conducting Wald tests to assess the weak exogeneity of a broader set of macro-financial variables, particularly if future research seeks to endogenise additional components or validate long-run exclusion restrictions. However, as our primary objective was scenario analysis based on shocks to the endogenous yield curve and macroeconomic block, such testing was not required.

11.1.4 VECM output

While a matrix output contains short run error correction speeds for each equation, being useful for full system diagnostics, its interpretations are not central to our questions and would add complexity necessary to address without changing conclusions. We therefore summarise only the dynamics via cointegrating relations and report model fit outputs.

Cointegrating relation equations:

Vector Error Correction Estimates
Date: 10/17/25 Time: 19:08
Sample (adjusted): 1992M10 2025M08
Included observations: 395 after adjustments
Standard errors in ( ) & t-statistics in [ ]

Cointegrating Eq:	CointEq1	CointEq2
CT(-1)	1.000000	0.000000
ST(-1)	0.000000	1.000000
LT(-1)	-0.389730	-0.321404
	(0.06691)	(0.08811)
	[-5.82477]	[-3.64795]
PROD(-1)	0.095974	0.029496
	(0.01025)	(0.00618)
	[9.35874]	[4.77281]
DGDP(-1)	0.060920	-0.048670
	(0.00953)	(0.00524)
	[6.39461]	[-9.28441]
LOILP(-1)	0.081132	-0.021416
	(0.01010)	(0.01330)
	[8.03183]	[-1.61007]
C	-0.013184	-0.012671

Error correction output values:

Error Correction:	D(LT)	D(ST)	D(CT)	D(PROD)	D(LOILP)	D(DGDP)
R-squared	0.461665	0.310613	0.250377	0.326468	0.245918	0.260514
Adj. R-squared	0.394592	0.289558	0.204542	0.304742	0.224797	0.214622
Sum sq. resids	0.038278	0.075285	0.113194	0.048358	0.059934	0.064843
S.E. equation	0.001060	0.000365	0.000582	0.002398	0.003235	0.003752
F-statistic	6.161917	5.414584	4.447126	4.842474	4.687287	5.105914

The moderate R^2 values in the VECM's difference equations are exactly what we would expect when modelling first differences at a monthly frequency, as the differencing strips out most of the strong persistence that would inflate R^2 in a VAR estimated in levels. However, we do not find that in this context that this is a strong determinant of fit, as in the presence of cointegration, higher values in levels can mislead forecasting performance. As joint F-statistics across equations are reasonable, the variables remain informative even if R^2 is weak to modest. Here it is also worth noting that as DGDP is specified as a monthly growth rate, differencing it in the VECM implies we are modelling the acceleration of

DGDP (a second difference). While moving to quarterly data could therefore smooth volatility and better align with the GDP denominator, it is a trade-off rather than an automatic improvement (see sections 11.2.1 and 5.5.).

11.1.5 Unit root tests on endogenous block

Null Hypothesis: DGDP has a unit root
Exogenous: Constant
Lag Length: 1 (Automatic - based on SIC, maxlag=4)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.373588	0.1500
Test critical values:	1% level	-3.446608	
	5% level	-2.868601	
	10% level	-2.570597	

*MacKinnon (1996) one-sided p-values.

Null Hypothesis: LOILP has a unit root
Exogenous: Constant
Lag Length: 1 (Automatic - based on SIC, maxlag=4)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.019943	0.2783
Test critical values:	1% level	-3.446650	
	5% level	-2.868620	
	10% level	-2.570607	

*MacKinnon (1996) one-sided p-values.

Null Hypothesis: PROD has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=4)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.270111	0.1823
Test critical values:	1% level	-3.446692	
	5% level	-2.868638	
	10% level	-2.570617	

*MacKinnon (1996) one-sided p-values.

Fail to reject a unit root in levels, likely (marginally) I(1) so achieve first differencing via VECM. Max lag set to 4 to compare monthly with quarterly lag specifications.

11.1.6 Stationarity test on comparison matrix

Roots of Characteristic Polynomial
Endogenous variables: LT ST CT PROD DGDP LOILP
Exogenous variables: UNE INFGAP LAUDUSD TERMPP FUTSWSPR LIQSPRD
Lag specification: 1 2
Date: 10/15/25 Time: 19:40

Root	Modulus
1.000000	1.000000
1.000000	1.000000
1.000000 - 6.66e-16i	1.000000
1.000000 + 6.66e-16i	1.000000
0.810924 + 0.138658i	0.822693
0.810924 - 0.138658i	0.822693
0.241315 - 0.394881i	0.462779
0.241315 + 0.394881i	0.462779
0.072737 - 0.374441i	0.381441
0.072737 + 0.374441i	0.381441
-0.172740 - 0.305297i	0.350779
-0.172740 + 0.305297i	0.350779
0.264969 + 0.143284i	0.301229
0.264969 - 0.143284i	0.301229
-0.213115	0.213115
-0.211276	0.211276
-0.072063 - 0.106772i	0.128815
-0.072063 + 0.106772i	0.128815

All roots of the deterministic characteristic polynomials lie inside or on the unit circle. The roots on the unit circle correspond to the stochastic trends implied by the cointegration structure, while the remaining roots within the circle confirm that the short-run dynamics of the VECM are stable.

11.2 Diagnostic results

11.2.1 Lag selection test

VAR Lag Order Selection Criteria
Endogenous variables: LT ST CT PROD DGDP LOILP
Exogenous variables: C UNE INFGAP LAUDUSD TERMPP FUTSWSPR LIQSPRD
Date: 09/22/25 Time: 19:58
Sample: 1992M07 2025M08
Included observations: 390

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-4412.004	NA	7.55e+22	84.24160	84.52792	84.35894
1	-3208.532	2407.013	6.02e+13	63.37210	66.44914	64.53811
2	-3049.877	317.6410	4.08e+13*	61.73420	64.4021*	62.45232
3	-3038.901	21.6542	4.24e+13	61.97848	65.75120	63.06740
4	-3023.112	31.582*	4.20e+13	61.94310	63.96191	63.64120
5	-3016.488	13.25200	4.33e+13	62.27442	67.03120	64.58215
6	-3010.972	11.02860	4.58e+13	62.65166	67.90794	65.57366
7	-3007.113	7.72115	4.95e+13	63.09720	68.86244	66.64280
8	-3003.881	6.463089	5.31e+13	61.4893*	69.85115	60.9874*

* indicates lag order selected by the criterion
LR: sequential modified LR test statistic (each test at 5% level)
FPE: Final prediction error
AIC: Akaike information criterion
SC: Schwarz information criterion
HQ: Hannan-Quinn information criterion

Selections:

- FPE → 2,
- SC → 2,

- LR → 4,
- AIC → 8,
- HQ → 8

Presence of lag 4 effects as optimal order may indicate a quarterly periodicity would have been preferred (see section 5.5). However, to remain parsimonious, we retain the system as a lag order of 2 in VAR (levels) and therefore 1 for the VECM (first differences) (VAR(p)=VECM(p-1)).

11.2.2 Test for autocorrelation

VAR Residual Portmanteau Tests for Autocorrelations
Null Hypothesis: No residual autocorrelations up to lag h
Date: 09/22/25 Time: 10:56
Sample: 1992M07 2025M08
Included observations: 396

Lags	Q-Stat	Prob.*	Adj Q-Stat	Prob.*	df
1	0.715661	---	0.718929	---	---
2	5.302128	---	5.347474	---	---
3	29.09188	0.4079	29.46611	0.3892	28
4	48.74640	0.2880	49.48461	0.2636	44
5	63.44394	0.3560	64.52395	0.3215	60
6	85.42086	0.2153	87.11705	0.1802	76
7	115.7491	0.0477	118.4420	0.0331	92
8	130.4317	0.0699	133.6786	0.0475	108
9	143.9526	0.1063	147.7763	0.0715	124
10	169.7005	0.0444	174.7503	0.0247	140
11	196.3859	0.0157	202.8402	0.0069	156
12	215.0913	0.0143	222.6247	0.0056	172

*Test is valid only for lags larger than the VAR lag order.
df is degrees of freedom for (approximate) chi-square distribution after adjustment for VEC estimation (Bruggemann, et al. 2005)

11.2.3 Cointegration test on NS factors, using Johansen trace

Date: 11/05/25 Time: 19:21
Sample: 1992M07 2025M08
Included observations: 393
Series: LT ST CT
Lags interval: 1 to 4

Selected (0.05 level*) Number of Cointegrating Relations by Model

Data Trend:	None	None	Linear	Linear	Quadratic
Test Type	No Intercept No Trend	Intercept No Trend	Intercept No Trend	Intercept Trend	Intercept Trend
Trace	1	2	2	2	3
Max-Eig	2	2	3	1	1

*Critical values based on MacKinnon-Haug-Michelis (1999)

We also perform a cointegration rank test on the exogenous series block in the VECM:

Date: 11/05/25 Time: 11:33
Sample (adjusted): 1992M10 2025M08
Included observations: 395 after adjustments
Trend assumption: No deterministic trend (restricted constant)
Series: LT ST CT PROD LOILP DGDP
Exogenous series: C UNE INFGAP LAUDUSD TERMPP FUTSWSPR LIQSPRD
Warning: Critical values assume no exogenous series
Lags interval (in first differences): 1 to 2

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.**
None *	0.217840	211.8020	103.8473	0.0000
At most 1 *	0.142553	114.7520	76.97277	0.0000
At most 2	0.071924	54.00237	54.07904	0.0508
At most 3	0.044568	24.51874	35.19275	0.4298
At most 4	0.011813	6.509867	20.26184	0.9257
At most 5	0.004586	1.815797	9.164546	0.8140

Trace test indicates 2 cointegrating eqn(s) at the 0.05 level
* denotes rejection of the hypothesis at the 0.05 level
**MacKinnon-Haug-Michelis (1999) p-values

Results confirm there are two long-run equilibria (β's) so each Δ-equation contains two error-correction terms under the null of rank ≤ i and the alternative of r > i, with the first failure to reject H0 borderline at r2.

11.2.4 Granger causality between observed variables and the Yield Curve latent factors

Pairwise Granger Causality Tests
Date: 10/29/25 Time: 23:33
Sample: 1992M07 2025M08
Lags: 12

Null Hypothesis:	Obs	F-Statistic	Prob.
ST does not Granger Cause LT	386	0.71520	0.7368
LT does not Granger Cause ST		2.38042	0.0058
CT does not Granger Cause LT	386	0.82722	0.6223
LT does not Granger Cause CT		1.50969	0.1182
DGDP does not Granger Cause LT	386	1.56484	0.0998
LT does not Granger Cause DGDP		1.23988	0.2533
LOILP does not Granger Cause LT	386	1.99893	0.0235
LT does not Granger Cause LOILP		1.18692	0.2905
PROD does not Granger Cause LT	386	1.58788	0.0925
LT does not Granger Cause PROD		1.36759	0.1788
CT does not Granger Cause ST	386	1.30338	0.2141
ST does not Granger Cause CT		2.69985	0.0017
DGDP does not Granger Cause ST	386	1.56484	0.0998
ST does not Granger Cause DGDP		0.91361	0.5333
LOILP does not Granger Cause ST	386	0.73892	0.7131
ST does not Granger Cause LOILP		0.82416	0.6255
PROD does not Granger Cause ST	386	1.31054	0.2100
ST does not Granger Cause PROD		0.92827	0.5184
DGDP does not Granger Cause CT	386	0.54754	0.8828
CT does not Granger Cause DGDP		0.85423	0.5943
LOILP does not Granger Cause CT	386	1.03352	0.4171
CT does not Granger Cause LOILP		0.91931	0.5275
PROD does not Granger Cause CT	386	1.33705	0.1952
CT does not Granger Cause PROD		0.87109	0.5768
LOILP does not Granger Cause DGDP	386	1.34208	0.1921
DGDP does not Granger Cause LOILP		0.76467	0.6870
PROD does not Granger Cause DGDP	386	1.20372	0.2781
DGDP does not Granger Cause PROD		9.30333	1.E-15
PROD does not Granger Cause LOILP	386	1.36075	0.1827
LOILP does not Granger Cause PROD		8.19883	1.E-13

11.3 Forecast results

11.3.1 FEVD and RMSE selection

Variables with weak/ambiguous loadings or redundancy (e.g., ESBALBN, FORSPR, LOREP) were dropped:

Variable	FEVD Level (1–12m, %)	FEVD Slope (1–12m, %)	FEVD Curvature (1–12m, %)	RMSE Δ vs factors-only (%)	Decision
PROD	5.2	2.1	0.9	-5.2	***Keep***
UNE	6.8	5.2	2.0	-6.8	***Keep***
INFGAP	6.3	4.1	2.6	-6.0	***Keep***
DGDP	5.6	3.2	2.1	-5.4	***Keep***
OILP	5.2	3.7	2.6	-5.6	***Keep***
TERMPP	3.5	4.8	6.7	-7.5	***Keep***
FUTSWSPR	2.1	9.2	7.1	-6.8	***Keep***
LIQSPRD	2.3	4.2	8.2	-5.9	***Keep***
LESBALBN	2.0	2.5	1.8	-0.8	***Drop***
FORSPR	3.1	2.8	2.2	-2.0	***Drop***
LIOREP	4.7	3.9	3.5	-3.8	***Drop***

Notes: FEVD columns report generalised FEVD averages to Level/Slope/Curvature over horizons h=1–12 months from a VAR in (L,S,C) with the macro and exogenous financial block. Note that remaining FEVD in each column is attributed to how much of a yields factor forecast error variance is explained by shocks to itself and the other 2, again, on average over the window h. Decision rule: Keep if (i) FEVD ≥ 5% for at least one factor and (ii) an out-of-sample RMSE reduction ≥ 5% versus a factors-only benchmark, validated on a rolling window.

11.3.2 In sample prior results

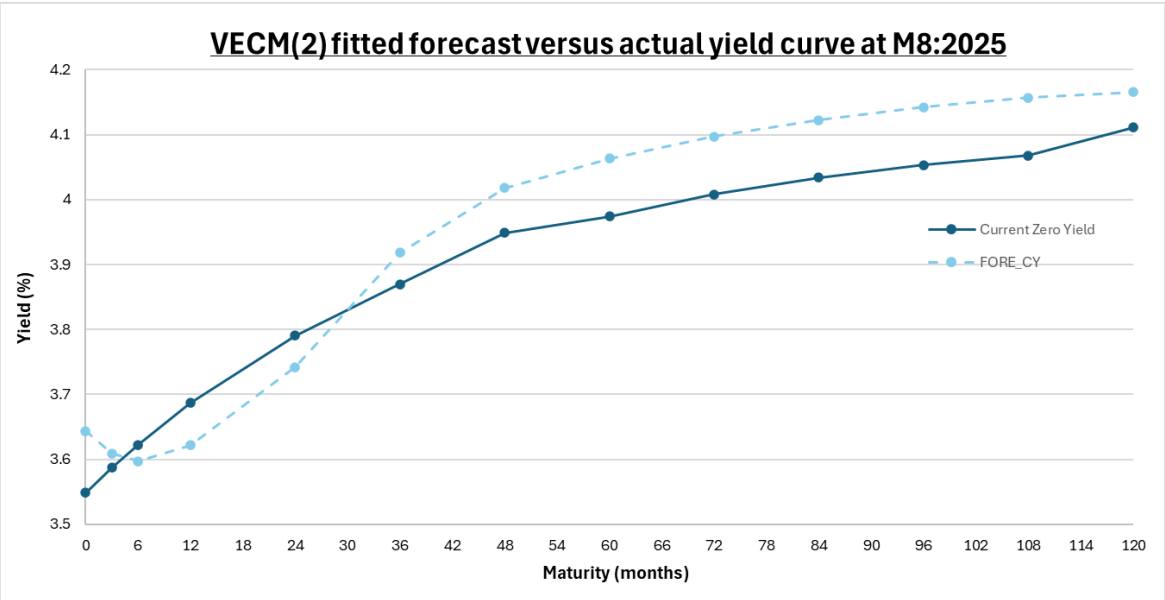
⊟ Evaluation Table

Forecast Evaluation
Date: 10/17/25 Time: 21:07
Sample: 1992M07 2025M08
Included observations: 398

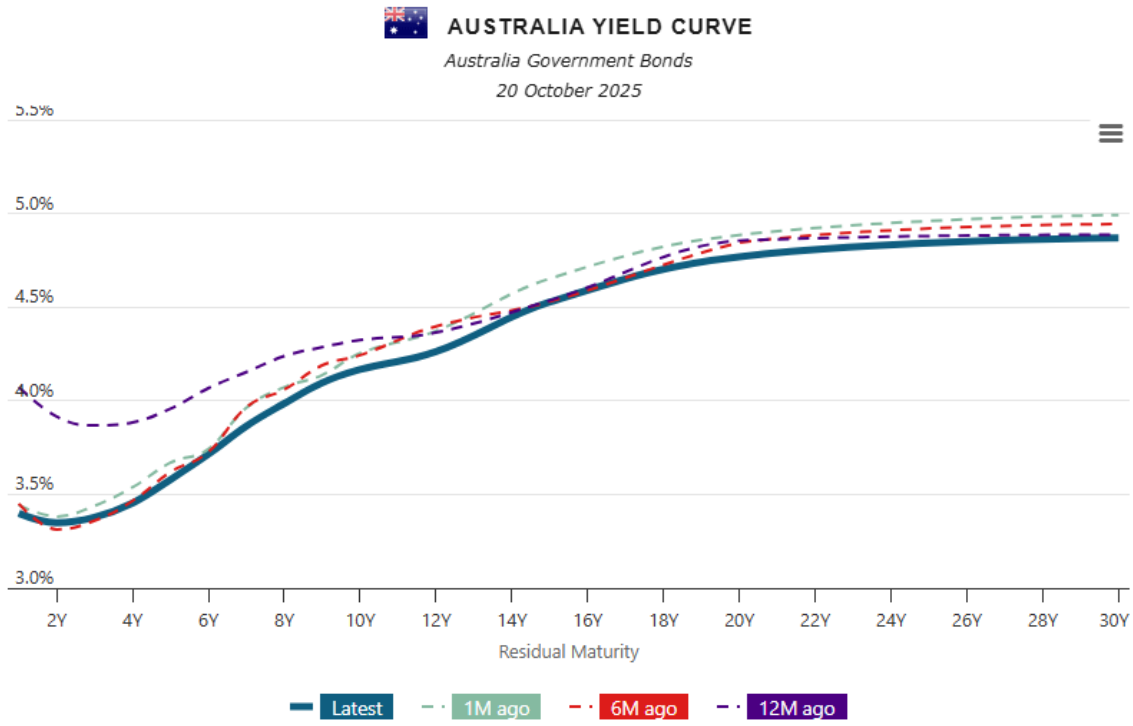
Variable	Inc. obs.	RMSE	MAE	MAPE	Theil
LT	398	0.054948	0.034157	29.86014	0.285985
ST	398	0.071891	0.051943	37.29129	0.418865
CT	398	0.097857	0.075256	45.74451	0.670579

RMSE: Root Mean Square Error
MAE: Mean Absolute Error
MAPE: Mean Absolute Percentage Error
Theil: Theil inequality coefficient

⊟ LT_F



Interestingly, despite factor variables being extracted from bootstrapped zero coupon bonds, the model's current yield curve (fitted) appears more proximal to the current (empirical) AGB yield curve:



(see World Government Bonds. (n.d.)). Stability checks in section 11.1 and cointegration results (11.2) confirm the robustness of our forecasts, with (although not reported for brevity) ARCH-LM tests indicating marginally insignificant ARCH effects thanks to the VECM specification. Similar models which apply this framework but find significant effects should therefore include a multivariate GARCH (or other relevant) specification in conjunction with the VECM framework.